

Monetary policy shocks and inflation in a monetary and fiscal policy mix regime

Diaz G. Sangeve^{*}, Sihlelelwe L. Nzima[†], Tshulani H. Khosa[‡]

Macroeconomics Group Project

May, 18 2026

^{*}**Study-Leader:** Department of Economics, Stellenbosch University. E-mail: 24991422@sun.ac.za.

[†]Department of Economics, Stellenbosch University. E-mail: 23854936@sun.ac.za.

[‡]Department of Economics, Stellenbosch University. E-mail: 25852094@sun.ac.za.

1. Introduction

The relationship between inflation and monetary policy is one of the most pressing macroeconomic issues. Global central banks have been operating on the assumption that inflationary pressure can be contained by rising interest rates since the early 1970s (Wansleben, 2023). The Taylor rule (1999) and the widespread use of inflation-targeting frameworks, initially investigated by Bernanke and Mishkin (1997), are based on this notion. However, this traditional perspective has been under scrutiny due to recent global developments, especially the aggressive post-2022 monetary tightening cycle (Hofmann, Manea and Mojon, 2024). The last fifteen years' experience indicates that contractionary monetary policy's ability to control inflation is neither universal nor independent of the larger policy environment. Rather, it seems to depend more and more on the position and limitations of fiscal policy.

Particularly in the framework of the fiscal theory of the price level (FTPL), a growing body of research emphasizes that inflation is fundamentally a mixed monetary and fiscal phenomena, while others maintain that it is either a completely monetary or fiscal phenomenon (Bianchi, Melosi and Rogantini-Picco, 2025; Weber and Feld, 2025; Khah and Ahmad, 2025). According to this approach, how fiscal authorities modify primary surpluses in response to shifts in public debt is crucial to how inflation responds to monetary policy shocks (Leeper, 1991; Sims, 1994; Leeper and Leith, 2016). Usually, there are two benchmark policy regimes. Fiscal policy passively stabilizes public debt through future surpluses in a monetary-dominated regime (Regime M), enabling monetary policy to successfully control inflation. On the other hand, in a fiscal-dominant regime (Regime F), inflation serves as the mechanism for stabilizing the real value of government obligations because fiscal policy is insufficiently adjusted.

This difference between fully supported and completely unbacked public debt is striking, even though it is useful for analysis. Fiscal authorities seldom function at these extremes in real life. The limits of governments' capacity to completely stabilize debt through future surpluses have been made clear by times of high public debt, such as those that followed the 2008 Global Financial Crisis (GFC) and the COVID-19 pandemic (Rawdanowicz, Turban, Haas, Crowe and Millot, 2021). However, fiscal systems do not completely give up on discipline. A more detailed description of policy interactions that permits partial fiscal support of public debt is motivated by this observation.

This idea is formalized by Bianchi, Faccini and Melosi (2023) through the notion of a *coexistence regime*, where backed and unbacked public debt coexist. According to this framework, inflation stabilizes a portion of government debt, while anticipated future

surpluses stabilize the other share. Therefore, the dynamics of inflation are jointly determined by monetary and fiscal authorities, with monetary policy staying active in relation to non-fiscal inflation and passive in relation to fiscal inflation. This regime captures an intermediate and arguably more realistic policy environment, particularly in economies facing fiscal constraints and rising debt service burdens.

The coexistence regime has important implications for how monetary policy shocks transmit to inflation. When public debt is only partially backed, an increase in interest rates raises debt service costs, which can worsen fiscal sustainability and amplify inflationary pressures through the debt service cost channel (Loewald, Steinbach and Rakgalakane, 2025). In such an environment, contractionary monetary policy may produce attenuated or even reversed effects on inflation relative to the standard New Keynesian (NK) benchmark.

This paper builds on this emerging literature by analysing the effects of a contractionary monetary policy shock in an environment where backed and unbacked public debt coexist. Specifically, it develops a NK-DSGE model that incorporates a fiscal rule with two components of public debt as shown by Bianchi *et al.* (2023). The analysis compares the transmission of monetary policy shocks across three regimes: the standard monetary-dominant regime (Regime M), the fiscal-dominant regime (Regime F), and the coexistence regime. Particular attention is given to the role of the debt service cost channel and how it shapes inflation and debt dynamics under different policy mixes.

This study makes three contributions. First, it expands the study of the transmission of monetary policy to a situation where public debt is partially unbacked. Second, it assesses how well the coexistence regime can balance the trade-off between debt sustainability and price stability. Third, it offers a quantitative evaluation of how monetary policy and fiscal capacity combine to affect macroeconomic outcomes, providing information pertinent to contemporary policy discussions.

2. The Model

The economy is populated by a representative household, a final-goods producing firm, a continuum of intermediate-goods producing firms indexed by $j \in [0, 1]$, and a government consisting of a fiscal authority and a monetary authority. Following [Bianchi *et al.* \(2023\)](#), the model features two parallel economies: an *actual economy* that tracks shocks affecting backed public debt and non-fiscal inflation, and a *shadow economy* that tracks shocks affecting unbacked public debt and fiscal inflation.

2.1. Households

Preferences. The representative household maximises the following expected lifetime utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} - \chi \frac{N_t^{1+\eta}}{1+\eta} \right], \quad (1)$$

where $\beta \in (0, 1)$ is the discount factor, $\sigma > 0$ is the inverse of the intertemporal elasticity of substitution (IES), $\eta > 0$ is the inverse of the Frisch elasticity of labour supply, and $\chi > 0$ is a labour-disutility scaling parameter. c_t and N_t denote real consumption and hours worked, respectively.

Budget constraint. In each period the household faces the nominal budget constraint:

$$(1 + \tau^c)P_t c_t + \frac{B_t}{R_t} = (1 - \tau_t^n)W_t N_t + B_{t-1} + \Psi_t, \quad (2)$$

where P_t is the price level, B_t is nominal holdings of one-period government bonds, R_t is the gross nominal interest rate, τ^c is a constant consumption tax rate, τ_t^n is the labour income tax rate, W_t is the nominal wage, and Ψ_t denotes nominal profits received from firms.

Dividing (2) by P_t and defining real quantities in lower-case letters ($b_t = B_t/P_t$, $w_t = W_t/P_t$, $\psi_t = \Psi_t/P_t$, $\pi_t = P_t/P_{t-1}$), the household's real budget constraint is presented as part of the dynamic programming formulation below.

Dynamic programming formulation. The household's sequential problem is cast in recursive form. The state vector at the beginning of period t is $\mathbf{s}_t = (b_{t-1}, Z_t, \tau_t^n)$, where b_{t-1} is real bond holdings carried from the previous period, Z_t is the aggregate technology level, and τ_t^n is the current labour income tax rate. The household chooses controls (c_t, N_t, b_t) each period. The **Bellman equation** is:

$$V(b_{t-1}, Z_t, \tau_t^n) = \max_{c_t, N_t, b_t} \left\{ \frac{c_t^{1-\sigma}}{1-\sigma} - \chi \frac{N_t^{1+\eta}}{1+\eta} + \beta \mathbb{E}_t[V(b_t, Z_{t+1}, \tau_{t+1}^n)] \right\}, \quad (3)$$

subject to the real budget constraint:

$$(1 + \tau^c) c_t + \frac{b_t}{R_t} = (1 - \tau_t^n) w_t N_t + \frac{b_{t-1}}{\pi_t} + \psi_t, \quad (4)$$

and a no-Ponzi condition $\lim_{T \rightarrow \infty} \beta^T \mathbb{E}_t[V(b_T, \cdot)] \geq 0$.

First-order conditions. Substituting the budget constraint (4) into (3) to eliminate c_t , the household optimises over N_t and b_t . Let $\mu_t \equiv V_b(b_{t-1}, Z_t, \tau_t^n)$ denote the marginal value of an additional unit of real bond wealth entering period t . By the envelope theorem:

$$\mu_t = V_b(b_{t-1}, Z_t, \tau_t^n) = \frac{c_t^{-\sigma}}{(1 + \tau^c) \pi_t}, \quad (5)$$

which equates the shadow value of inherited wealth to the marginal utility of the additional consumption it finances, deflated by the consumption tax wedge and gross inflation.

FOC for bonds (b_t):

Differentiating (3) with respect to b_t and applying the envelope condition (5) one period forward:

$$\frac{\partial}{\partial b_t} : \quad -\frac{c_t^{-\sigma}}{(1 + \tau^c)} \cdot \frac{1}{R_t} = -\beta \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{(1 + \tau^c) \pi_{t+1}} \right]. \quad (6)$$

Cancelling the common factor $(1 + \tau^c)^{-1}$ and rearranging yields the standard **Euler equation**:

$$\boxed{\frac{1}{R_t} = \beta \mathbb{E}_t \left[\left(\frac{c_t}{c_{t+1}} \right)^\sigma \frac{1}{\pi_{t+1}} \right]}. \quad (7)$$

This condition equates the marginal cost of purchasing one additional bond (the foregone consumption of $1/R_t$ units today) to the expected discounted marginal benefit of the real wealth it delivers next period (after deflation by π_{t+1}). Intuitively, if R_t rises, the bond becomes cheaper, and the household saves more—the standard intertemporal substitution channel.

FOC for labour (N_t):[§]

[§]For the remainder of the paper, labour input is measured as the working-age-population—individuals aged 16-64.

Differentiating (3) with respect to N_t :

$$\frac{\partial}{\partial N_t} : \quad -\chi N_t^\eta + \frac{c_t^{-\sigma}}{(1 + \tau^c)} (1 - \tau_t^n) w_t = 0. \quad (8)$$

Rearranging:

$$\boxed{\chi N_t^\eta c_t^\sigma = \frac{(1 - \tau_t^n)}{(1 + \tau^c)} w_t.} \quad (9)$$

This condition equates the household's marginal rate of substitution between leisure and consumption—the left-hand side—to the after-tax real wage on the right-hand side. A higher consumption tax $(1 + \tau^c)$ or a higher labour tax $(1 - \tau_t^n)$ drives a wedge between the pre-tax wage and the household's effective return to working.

Transversality condition.

For the optimum to be interior and the budget constraint to bind in the limit, the following transversality condition must hold:

$$\lim_{T \rightarrow \infty} \beta^T \mathbb{E}_t \left[\frac{c_T^{-\sigma}}{(1 + \tau^c)} \frac{b_T}{\pi_T} \right] = 0, \quad (10)$$

ruling out explosive debt paths sustained by ever-growing borrowing.

2.2. The Production Sector

The production sector consists of two tiers: a perfectly competitive final-goods firm that bundles differentiated intermediate goods into a single consumption good, and a continuum of monopolistically competitive intermediate-goods firms indexed by $j \in [0, 1]$, each of which produces a distinct variety and sets its own price subject to nominal rigidity.

Variables. The key variables of the production sector are as follows. Y_t denotes aggregate real output of the final good in period t . $Y_t(j)$ is the quantity of intermediate variety j demanded by the final-goods firm. P_t is the aggregate price level (the CES price index dual to the aggregator). $P_t(j)$ is the nominal price charged by intermediate firm j . P_t^* is the common optimal reset price chosen by all re-optimising firms in period t ; because the model is symmetric, every firm that re-optimises in the same period sets the same price. $N_t(j)$ is hours of labour hired by intermediate firm j , and $N_t = \int_0^1 N_t(j) dj$ is aggregate labour demand. Z_t is the aggregate technology level, common to all intermediate producers. $mc_t = w_t/Z_t$ is the real marginal cost of production, equal to the real wage divided by the marginal product of labour. $d_t(j)$ denotes the real profits of intermediate firm j in period t . $g_{1,t}$ and $g_{2,t}$ are auxiliary recursive variables that summarise, respectively, the discounted

stream of future nominal revenues weighted by marginal cost, and the discounted stream of future nominal revenues; they are introduced to render the infinite-horizon optimal pricing condition computationally tractable.

Parameters. $v > 1$ is the constant elasticity of substitution across intermediate varieties in the CES aggregator. It controls the degree of monopoly power: a higher v implies that varieties are closer substitutes, eroding mark-up power, with the steady-state mark-up equal to $v/(v - 1)$. $\theta \in [0, 1)$ is the Calvo price-stickiness parameter: in each period, a fraction θ of intermediate firms cannot adjust their price, while the remaining fraction $(1 - \theta)$ re-optimize freely. The slope of the New Keynesian Phillips curve $\kappa = (1 - \theta)(1 - \beta\theta)/\theta$ is a function of both θ and the household discount factor β ; a higher θ (stickier prices) flattens the Phillips curve, reducing the inflation response to changes in marginal cost. $\rho_z \in (0, 1)$ is the AR(1) persistence parameter of the technology shock. σ_z^2 is the variance of the innovation ξ_t^z to the technology process.

Final-goods firm. The final-goods firm aggregates intermediate varieties via the CES technology and, from profit maximisation, the demand schedule for variety j and the aggregate price index are:

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-v} Y_t, \quad (11)$$

$$P_t = \left[\int_0^1 P_t(j)^{1-v} dj \right]^{\frac{1}{1-v}}. \quad (12)$$

Intermediate-goods firms. Each intermediate firm employs labour under a linear technology and faces a real marginal cost given by:

$$Y_t(j) = Z_t N_t(j), \quad (13)$$

$$mc_t = \frac{w_t}{Z_t}, \quad (14)$$

$$\ln Z_t = \rho_z \ln Z_{t-1} + \xi_t^z, \quad \xi_t^z \sim \mathcal{N}(0, \sigma_z^2). \quad (15)$$

Calvo price setting. Each period a fraction θ of firms maintain last period's price; the remaining fraction $(1 - \theta)$ re-optimize. The aggregate price index therefore evolves as:

$$P_t(j)^{1-v} = \theta P_{t-1}(j)^{1-v} + (1 - \theta) (P_t^*)(j)^{1-v}. \quad (16)$$

The optimal reset price and its recursive representation in terms of the auxiliary variables

$g_{1,t}$ and $g_{2,t}$ are:

$$\frac{P_t^*}{P_t} = \frac{v}{v-1} \cdot \frac{g_{1,t}}{g_{2,t}}, \quad (17)$$

$$g_{1,t} = Y_t mc_t c_t^{-\sigma} + \beta\theta \mathbb{E}_t \left[\pi_{t+1}^v g_{1,t+1} \right], \quad (18)$$

$$g_{2,t} = Y_t c_t^{-\sigma} + \beta\theta \mathbb{E}_t \left[\pi_{t+1}^{v-1} g_{2,t+1} \right]. \quad (19)$$

The reset price is a mark-up $v/(v-1)$ over a forward-looking weighted average of current and expected future real marginal costs. $g_{1,t}$ accumulates the discounted marginal-cost-weighted revenue stream that a firm locked into today's price would earn, while $g_{2,t}$ accumulates the corresponding revenue stream absent the marginal-cost weight; their ratio therefore yields the relevant average cost measure. Both auxiliary variables discount future payoffs by $\beta\theta$, reflecting time preference (β) and the probability of remaining price-locked (θ), and inflate future values by π_{t+1}^v (or π_{t+1}^{v-1}) to account for the erosion of real revenues from a fixed nominal price as the price level rises.

2.3. Government

Government budget constraint. The government's nominal flow budget constraint is:

$$\frac{B_t}{R_t} + T_t = B_{t-1} + G_t, \quad (20)$$

where T_t denotes nominal tax revenue and G_t nominal government expenditure. Dividing by P_t , the real budget constraint is:

$$\frac{b_t}{R_t} + \tau_t = \frac{b_{t-1}}{\pi_t} + g_t, \quad (21)$$

and the law of motion for real public debt is:

$$\boxed{b_t = \frac{R_{t-1}}{\pi_t} b_{t-1} - s_t}, \quad (22)$$

where $s_t = \tau_t - g_t$ is the real primary surplus. To explicitly track the burden of existing obligations, we define the net debt service cost, dsc_t , as:

$$dsc_t = \left(\frac{R_{t-1}}{\pi_t} - 1 \right) b_{t-1}. \quad (23)$$

A contractionary monetary policy shock raises R_t , which, for a given inflation rate, mechanically increases the debt service cost (dsc_t) and, consequently, public debt. The

quantitative importance of this channel is proportional to the outstanding debt stock.

2.3.1 Coexistence Regime

Following Bianchi *et al.* (2023), total real public debt is decomposed into a *backed* component b_t^m and an *unbacked* component b_t^f :

$$b_t = b_t^m + b_t^f, \quad b_t^m \equiv b_t - b_t^f. \quad (24)$$

Aggregate gross inflation decomposes multiplicatively into *non-fiscal inflation* Π_t^m (backed debt) and *fiscal inflation* Π_t^f (unbacked debt):

$$\Pi_t = \Pi_t^m \Pi_t^f, \quad \Pi_t^m \equiv \frac{\Pi_t}{\Pi_t^f} \quad (25)$$

Fiscal rule (coexistence regime). The fiscal authority adjusts the primary surplus-to-GDP ratio according to:

$$\frac{s_t}{y_t} = \left(\frac{s}{y} \right) \left(\frac{b_{t-1}/(R_{t-1}y_{t-1})}{b_{t-1}^f/(R_{t-1}y_{t-1})} \right)^{\delta^{PF}} \left(\frac{b_{t-1}^f/(R_{t-1}y_{t-1})}{b/(Ry)} \right)^{\delta^{AF}}, \quad (26)$$

where variables without time subscripts denote steady-state values. The parameter δ^{PF} governs the passive fiscal response to backed debt (Regime M behaviour), while δ^{AF} governs the active fiscal response to unbacked debt (Regime F behaviour). The conditions for a unique bounded equilibrium require δ^{PF} to be sufficiently large (passive fiscal: $\beta^{-1} - \delta^{PF} < 1$) and δ^{AF} to be close to zero (active fiscal: $\beta^{-1} - \delta^{AF} > 1$).

Monetary rule (coexistence regime). The monetary authority follows a Taylor-type interest rate rule with two inflation components:

$$\frac{1}{R_t} = \frac{1}{R} \left(\frac{1/R_{t-1}}{1/R} \right)^{\rho_R} \left[\left(\frac{1/\Pi_t}{1/\Pi_t^f} \right)^{\alpha_\pi^{AM}} \left(\frac{1/\Pi_t^f}{1/\Pi} \right)^{\alpha_\pi^{PM}} \right]^{1-\rho_R} e^{\varepsilon_t^R}, \quad (27)$$

where $\rho_R \in (0, 1)$ is the interest-rate-smoothing parameter, α_π^{AM} is the active monetary response to non-fiscal (backed) inflation, α_π^{PM} is the passive monetary response to fiscal inflation, and ε_t^R is an i.i.d. monetary policy shock with mean zero and variance σ_R^2 . The Taylor principle requires $\alpha_\pi^{AM}\beta > 1$ (active monetary policy vis-à-vis non-fiscal inflation) and $\alpha_\pi^{PM}\beta < 1$ (passive monetary policy vis-à-vis fiscal inflation).

Linearising (27) gives:

$$\hat{r}_t = \rho_R \hat{r}_{t-1} + (1 - \rho_R) \left[\alpha_\pi^{AM} (\hat{\pi}_t - \hat{\pi}_t^f) + \alpha_\pi^{PM} \hat{\pi}_t^f \right] + \varepsilon_t^R, \quad (28)$$

where hatted variables denote log-deviations from steady state.

2.3.2 Standard Regimes M and F

For comparison, under the standard regimes the fiscal rule is:

$$\frac{s_t}{y_t} = \left(\frac{s}{y} \right) \left(\frac{b_{t-1}/(R_{t-1}y_{t-1})}{b/(Ry)} \right)^\delta, \quad (29)$$

and the monetary rule is:

$$\frac{1}{R_t} = \frac{1}{R} \left(\frac{1/R_{t-1}}{1/R} \right)^{\rho_R} \left(\frac{1/\Pi_t}{1/\Pi} \right)^{\alpha_\pi(1-\rho_R)} e^{\varepsilon_t^R}. \quad (30)$$

Regime M obtains when $\alpha_\pi\beta > 1$ and $\beta^{-1} - \delta < 1$. **Regime F** obtains when $\alpha_\pi\beta < 1$ and $\beta^{-1} - \delta > 1$.

2.3.3 Shadow Economy

Following Bianchi *et al.* (2023), the model appends a shadow economy that mirrors the actual economy but tracks only unbacked debt b_t^f and fiscal inflation Π_t^f . The shadow economy's fiscal and monetary rules are:

$$\frac{s_t^f}{y_t^f} = \left(\frac{s}{y} \right) \left(\frac{b_{t-1}^f/(R_{t-1}^f y_{t-1}^f)}{b/(Ry)} \right)^{\delta^{AF}}, \quad (31)$$

$$\frac{1}{R_t^f} = \frac{1}{R} \left(\frac{1/R_{t-1}^f}{1/R} \right)^{\rho_R} \left(\frac{1/\Pi_t^f}{1/\Pi} \right)^{\alpha_\pi^{PM}(1-\rho_R)} e^{\zeta \varepsilon_t^R}, \quad (32)$$

alongside the shadow net debt service cost:

$$dsc_t^f = \left(\frac{R_{t-1}^f}{\Pi_t^f} - 1 \right) b_{t-1}^f, \quad (33)$$

where $\zeta \in (0, 1)$ is the fraction of the monetary policy shock that influences unbacked public debt. In the shadow economy, fiscal policy is active ($\delta^{AF} = 0$) and monetary policy is passive ($\alpha_\pi^{PM}\beta < 1$), i.e. Regime F applies. All other equilibrium conditions in the shadow economy are identical to those of the actual economy.

2.4. Market Clearing and Aggregation

The goods market clears when:

$$y_t = c_t + g_t. \quad (34)$$

Total real public debt and gross inflation are the sums of their backed and unbacked components:

$$b_t = b_t^f + b_t^m,$$

while aggregate gross inflation decomposes multiplicatively:

$$\Pi_t = \Pi_t^f \Pi_t^m, \quad \Pi_t^m \equiv \frac{\Pi_t}{\Pi_t^f}.$$

3. Steady State

In the non-stochastic steady state all shocks are set to zero. We normalise gross aggregate inflation to unity, $\Pi = 1$ (equivalently, the net inflation rate is zero), and denote the steady-state values of variables without time subscripts[¶].

Household. The Euler equation (7) in steady state gives:

$$R = \frac{1}{\beta}. \quad (35)$$

This implies an annual steady-state nominal interest rate of approximately 4% when $\beta = 0.99$.

Labour market. In steady state, using the labour supply condition (9) with constant tax rates:

$$w = \frac{(1 + \tau^c)}{(1 - \tau^n)} \chi N^\eta c^\sigma. \quad (36)$$

Firms. With zero steady-state inflation, the optimal price reset condition (17) collapses to $P^*/P = 1$, and the price mark-up condition gives:

$$mc = \frac{v - 1}{v}. \quad (37)$$

[¶]Complete system of Nonlinear Equilibrium Equations and Steady State Conditions found in [Appendix A](#) and [Table B.1](#).

Since $mc = w/Z$ and we normalise $Z = 1$:

$$w = \frac{v-1}{v}. \quad (38)$$

Technology. The production function (13) in steady state gives $Y = ZN = N$ (with $Z = 1$).

Government. The government budget constraint (22) in steady state yields:

$$s = \left(\frac{1}{\beta} - 1 \right) b = (\beta^{-1} - 1) b. \quad (39)$$

Consequently, the steady-state debt service cost equals the primary surplus:

$$dsc = \left(\frac{R}{\Pi} - 1 \right) b. \quad (40)$$

The primary surplus-to-GDP ratio is:

$$\frac{s}{y} = (\beta^{-1} - 1) \frac{b}{y}. \quad (41)$$

Goods market. $y = c + g$, giving $c/y = 1 - g/y$.

Labour disutility. Combining (36) and (38) and the steady-state goods-market condition, the labour disutility parameter χ is pinned down by:

$$\chi = \frac{(1 - \tau^n)}{(1 + \tau^c)} \frac{(v-1)/v}{N^\eta c^\sigma}. \quad (42)$$

4. Calibration

The model is calibrated on a quarterly basis for the United States, spanning the first quarter of 1982 through the third quarter of 2025. Data availability determines this end date since all aggregate variables must be transformed into per capita figures. This transformation relies on the OECD working age population series for the US available on FRED under the identifier LFWA64TTUSM647S for ages 15 to 64. This specific series is only published up to September 2025. Beginning the sample in the first quarter of 1982 isolates the monetary policy regime following the Volcker disinflation and prevents the high inflation period of the late 1970s from distorting the calculated moments.

We assign standard production and preference parameters based on widely accepted values within the New Keynesian DSGE framework. The household discount factor remains fixed at $\beta = 0.99$ (Rotemberg and Woodford, 1997). This choice yields an annualised gross real return of approximately four per cent in the steady state, which closely aligns with the average federal funds rate throughout our timeframe. We define the inverse intertemporal elasticity of substitution as $\sigma = 2$ (Aruoba *et al.*, 2018) and the inverse Frisch elasticity as $\eta = 1$ (Davig and Leeper, 2011). Guided by Galí (2015) and Chen *et al.* (2022), the elasticity of substitution among intermediate goods is $v = 9$, and the Calvo price stickiness parameter is $\theta = 0.75$. These combined settings generate a steady state price markup of 12.5% alongside an average price duration of four quarters. Gross inflation in the steady state is normalised to one. Rather than selecting the labour disutility weight independently, we derive $\chi = 1.18$ residually to ensure the intratemporal first-order condition of the household holds at the resulting steady state.

Instead of borrowing figures from external literature, we calibrate two fiscal ratios using the average values from our constructed dataset. The public debt to output ratio is set to $b/y = 0.746$, representing the average US federal debt held by the public divided by nominal GDP from the first quarter of 1982 through the third quarter of 2025. Similarly, the government spending to output ratio is established at $g/y = 0.23$, calculated straight from the sample average of G/Y provided by FRED. While these parameters fall within standard bounds for similar DSGE models, anchoring the steady state to our specific empirical window ensures internal consistency between the calibrated economy and the statistical moments detailed in Section 5. The consumption tax rate is specified as $\tau^c = 0.023$ to match the effective average rate found in US records. We then find the steady state labour income tax rate τ^n residually through the government budget constraint, adopting the approach utilised by Leeper *et al.* (2017) and Bianchi *et al.* (2023).

We select policy reaction coefficients to guarantee a unique bounded equilibrium for every

regime according to the stipulations outlined in Appendix B. For Regime M, we apply a standard Taylor rule response of $\alpha_\pi = 1.5$ (Smets and Wouters, 2007) paired with a fiscal feedback parameter of $\delta = 1.2$. This combination securely positions fiscal policy in the passive zone. Regime F features a symmetric calibration where $\alpha_\pi = 0.5$ and $\delta = 0$, reflecting the choices of Leeper and Leith (2016) and Bianchi *et al.* (2023). The coexistence regime adopts these precise values for the respective active and passive elements. This results in $\alpha_\pi^{AM} = 1.5$ and $\delta^{PF} = 1.2$ for the Regime M portion, alongside $\alpha_\pi^{PM} = 0.5$ and $\delta^{AF} = 0$ for the Regime F portion. We establish the baseline share of the monetary policy shock applied to unbacked debt at $\zeta = 0.75$. This parameterisation reflects a macroeconomic climate characterised by high public debt and limited fiscal space, meaning that primary surpluses fail to absorb a large fraction of interest rate shocks. For context, Smets and Wouters (2024) calculate US fiscal backing to be approximately 0.83, though their stock measurement does not perfectly align with our shock loading parameter ζ . We explore the model robustness under different ζ assignments in the Sensitivity section.

We independently calibrate the autoregressive coefficients governing the exogenous processes. The monetary policy shock carries a persistence parameter of $\rho_R = 0.85$, designed to reflect the gradual easing of the federal funds rate following the post-pandemic tightening phase. We assign the technology shock a persistence of $\rho_z = 0.9$, which represents a conventional baseline in the empirical DSGE literature (Smets and Wouters, 2003).

5. Data

This section evaluates the model’s ability to replicate United States business cycle dynamics over the sample period from 1982:Q1 to 2025:Q3. The sample period concludes in the third quarter of 2025, rather than the fourth, due to data availability constraints regarding the US working-age population series on the FRED database. To generate the theoretical moments for Regimes M, F, and the coexistence framework, the dynamic systems are solved using a first-order Taylor approximation around their respective non-stochastic steady states. The simulated business cycle statistics are then directly extracted from the asymptotic variance-covariance matrix of the state-space representation and are Hodrick-Prescott (HP) filtered ($\lambda = 1600$) to ensure a methodologically consistent comparison with the de-trended empirical data. Table 1 presents the resulting standard deviations of key macroeconomic variables and their contemporaneous correlations with output, contrasting the model’s theoretical predictions against the empirical observations.

Table 1: Business Cycle Properties: Model versus Data

Variable	Standard Deviation (%)				Correlation with Output			
	Data	Reg M	Reg F	Coexist	Data	Reg M	Reg F	Coexist
Output	1.244	2.233	0.690	0.794	1.000	1.000	1.000	1.000
Consumption	1.236	2.901	0.896	1.031	0.884	1.000	1.000	1.000
Inflation (CPI)	0.536	1.322	2.395	1.624	0.344	0.987	0.716	0.193
Public debt	2.552	83.115	1.677	21.579	-0.356	-0.226	0.997	0.091
Fed funds rate	0.568	1.044	2.629	2.175	0.282	-0.989	0.547	-0.264

Notes: The model’s second moments are derived by simulating the model with a 1% technology shock. For the data, aggregate variables are in real terms (FRED: GDPDEF for the GDP deflator; GDP for output; PCECC96 for household consumption; GFDEBTN for public debt), converted to per capita terms using the US working-age population (FRED: LFWA64TTUSM647S), transformed into natural logarithm, and de-trended using the Hodrick–Prescott filter ($\lambda = 1600$). The inflation rate (FRED: CPIAUCSL) and nominal interest rate (FRED: FEDFUNDS) are expressed in gross terms and are not log-transformed. All data for the period 1982:Q1–2025:Q3 are obtained from FRED (<https://fred.stlouisfed.org/>).

The simulated second moments reveal a trade-off between price stability, output stability, and debt stabilisation across the three regimes. Under Regime M, active monetary policy anchors inflation, yielding an inflation standard deviation (1.32%) closest to the empirical US value (0.54%). Because the price level is firmly anchored, real quantities must absorb most of the technology shock, generating the highest output volatility in the model (2.23%, against 1.24% in the data) and substantially exceeding the corresponding figures under Regime F (0.69%) and the coexistence regime (0.79%). The accompanying debt volatility of 83.12% is not the result of debt “absorbing” the shock but of near-unit-root debt dynamics under passive fiscal policy: passivity ($\beta^{-1} - \delta^{PF} < 1$) ensures long-run solvency but allows transitory shocks to produce large, persistent deviations in b_t before

mean-reversion. Regime F achieves highly stable debt dynamics (1.68%) by the opposite route: the price level adjusts to revalue the real stock of unbacked debt in line with the expected discounted stream of future surpluses, generating pronounced “fiscal inflation” (2.40%, roughly 4.5 times the empirical value) while dampening output fluctuations. The coexistence regime, by structurally splitting the debt stock, sits between these extremes, limiting debt volatility to 21.58%, roughly one-quarter the Regime M figure, while keeping inflation volatility (1.62%) only modestly above Regime M.

Regarding correlation with output, US data show public debt to be mildly countercyclical (-0.356), a sign and rough magnitude qualitatively reproduced only by Regime M (-0.226), where primary surpluses rise during expansions and reduce the real backed-debt stock. Regime F generates a near-perfect positive correlation (0.997) via the discounted-surplus-valuation mechanism characteristic of FTPL economies. With active fiscal policy fixing s_t/y_t in this regime, expected future real surpluses scale with output, so the real value of unbacked debt, pinned down by the expected present value of those surpluses, moves positively with y_t by construction. The coexistence regime blends both channels to produce a near-zero correlation (0.091). On this moment alone, Regime M is in fact closer to the data than the coexistence regime in absolute terms (distance 0.130 versus 0.447).

The perfect correlation between consumption and output (1.000) across all three regimes, against an empirical value of 0.884, is a mechanical consequence of using only a technology shock. The positive inflation–output correlations produced by every regime (0.987, 0.716, 0.193) are qualitatively consistent with the empirical sign (0.344), even though the textbook transitory-technology argument predicts the opposite sign through the marginal-cost channel. The calibrated persistence $\rho_z = 0.9$ partially reconciles this, where a near-permanent productivity improvement raises expected future consumption, and through the consumption–Euler relation the household raises current consumption demand sufficiently to push current real marginal cost.

The cyclical behaviour of the nominal interest rate is shaped jointly by regime-specific monetary feedback and the limited shock structure of this simulation. In the data, the federal funds rate is mildly procyclical (0.282). Regime M generates a strongly negative rate–output correlation (-0.989), while under Regime F and the coexistence regime, where monetary policy is passive in relation to fiscal inflation ($\alpha_\pi^{PM}\beta < 1$) and therefore fails the Taylor principle, the rate–output correlation rises to 0.547 and -0.264 respectively.

Taken together, no single regime dominates on every moment, but the coexistence framework offers the most balanced overall representation of US business cycle properties without the explosive variances of Regime M or the counterfactual debt–output comovement of Regime F.

6. Results

Figure 1 presents the impulse response functions (IRFs) to a one percentage point contractionary monetary policy shock under three policy regimes: Regime M, Regime F, and the Coexistence regime. The responses across regimes are vary and reveal that the macroeconomic transmission of monetary policy is mostly shaped by the fiscal environment in which it operates.

6.1. Regime M

The contractionary shock generates dynamics that are generally consistent with the standard New Keynesian transmission mechanism under monetary dominance, where the central bank aggressively adjusts interest rates in response to inflation while the fiscal authority endogenously consolidates to stabilise debt. The intertemporal substitution effect causes consumption to continuously drop after the shock because higher real interest rates increase the opportunity cost of current expenditure. In addition, when debt increases, the passive fiscal authority tightens, creating a negative wealth impact that further stifles private demand. Real earnings and hours worked both significantly decline when the labour market deteriorates, reflecting the decline in aggregate demand and the diminished motivation for businesses to employ and create.

As seen by the drop in inflation after the shock, an active monetary authority may successfully anchor expectations and accomplish disinflation by raising the real interest rate enough above inflation. By increasing the primary surplus, fiscal policy supports this adjustment process and strengthens rather than lessens the contractionary impacts of monetary tightening. Fiscal consolidation eventually stabilizes debt dynamics, limiting persistent debt build up over the medium to long term, even while the debt-to-GDP ratio initially rises owing to decreased production and increased debt-servicing costs.

Regime M's main consequence is that monetary policy works in the traditional sense: the shock spreads smoothly along the real rate channel, lowering inflation and economic activity as planned. But there are costs associated with this achievement. The fiscal tightening that maintains the regime's price stability features adds a procyclical drag that intensifies the recession, and the drops in employment, real earnings, and consumption constitute a significant short-term loss.

6.2. Regime F

Regime F is characterised by a perverse inflation response: inflation rises persistently following monetary tightening, contrary to the conventional monetary transmission mechanism.

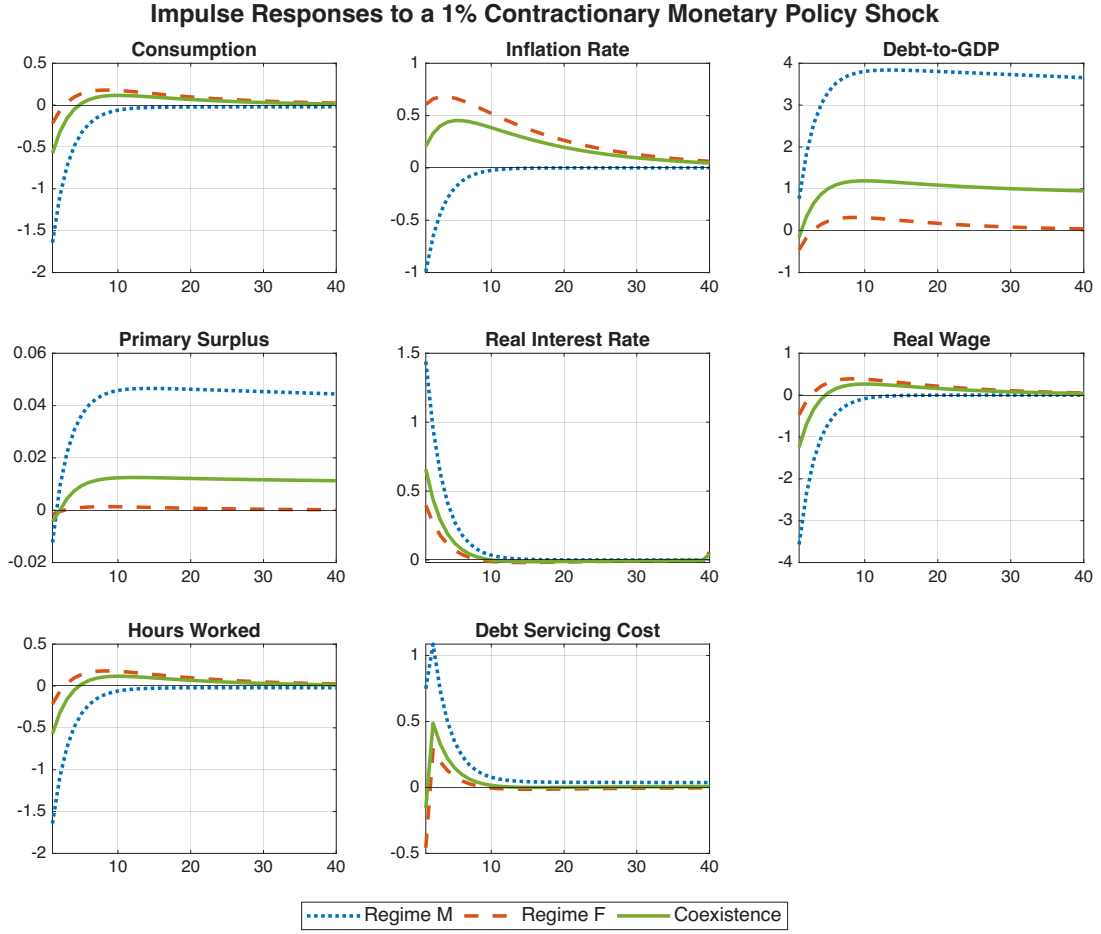


Figure 1: Impulse Responses to a Contractionary Monetary Policy Shock. Coexistence regime (green solid line); Regime F (red dashed line); and Regime M (blue dotted line). The variables are expressed as percentage deviations from steady state, except for the fiscal variables, which are expressed as linear deviations from steady state.

This outcome reflects the Fiscal Theory of the Price Level (FTPL). In the absence of fiscal adjustment, higher interest rates increase the government's debt-servicing burden, and the intertemporal budget constraint is restored through a higher price level. Consequently, agents anticipate future inflation, causing inflation to rise on impact. Monetary tightening therefore becomes counter-productive.

This point of view is supported by the budgetary reactions. The debt-to-GDP ratio is rising steadily and significantly, while the primary surplus is still mainly unresponsive. Increased interest rates create a trajectory of declining fiscal sustainability by compounding the expenses of debt servicing. A regime of macroeconomic fragility when monetary policy loses its ability to stabilize the economy is indicated by mounting debt, rising inflation, and poor fiscal response.

As a result, the actual economy declines. Compared to Regime M, real earnings and

hours worked are declining more slowly, which reflects both the reduction in demand and the depreciation of buying power brought on by inflation. Expectations of inflationary debt reduction only cause a transient spike in consumption, which then declines as real earnings decline. The lack of fiscal adjustment keeps the real interest rate from producing disinflationary gains, even though it is still high.

6.3. Coexistence Regime

The Coexistence regime, in which both monetary and fiscal authorities play active stabilising roles, generates outcomes that are more balanced than those observed under either pure monetary or fiscal dominance. Its key advantage lies in simultaneously achieving disinflation and limiting macroeconomic instability.

After the contractionary shock, inflation decreases, demonstrating the continued efficacy of the monetary transmission mechanism. Monetary policy is still able to stabilize inflation expectations, in contrast to Regime F. However, because partial fiscal accommodation cushions aggregate demand without completely countering the contractionary monetary impulse, the consumption decrease is far less severe than it was under Regime M. Additionally, real earnings and hours worked decline less and bounce back faster, suggesting less job losses and income losses.

Although much less consistently than under Regime F, the debt-to-GDP ratio rises in the short term before stabilizing as fiscal adjustment progressively takes effect. This intermediate debt trajectory is indicative of a well-balanced policy framework where fiscal policy maintains debt sustainability while assisting in stabilization. Although debt-servicing expenses increase somewhat, they do not produce the explosive dynamics seen in Regime F.

Overall compared to Regimes M and F, the Coexistence regime more successfully spreads the cost of macroeconomic stabilization. By providing relatively small fiscal support, it reduces the recessionary costs of adjustment while maintaining the disinflationary role of monetary policy. As a result, it produces a more stable macroeconomic result overall by avoiding both the deeper recession associated with tight monetary dominance and the inflationary instability associated with fiscal dominance.

6.4. Findings

A key finding of the literature on monetary-fiscal interaction is reinforced by the comparison of the three regimes: the fiscal environment in which monetary policy operates is crucial to its effectiveness (Leeper, 1991). Regime M demonstrates that monetary policy may lead

to price stability, but only at a substantial real economic cost and with supporting fiscal adjustment. The constraints of monetary policy under fiscal dominance are shown by Regime F, where tightening becomes ineffective due to rising inflation, debt build up, and macroeconomic instability. In contrast, the Coexistence regime shows that coordinated policy may lower the actual costs of adjustment and promote disinflation at the same time.

These findings carry important policy implications. In economies characterised by elevated debt levels and limited fiscal space, monetary tightening alone may be insufficient to stabilise inflation and can potentially worsen fiscal vulnerabilities. Effective macroeconomic stabilisation therefore requires credible fiscal adjustment alongside monetary restraint. Without such coordination, the economy risks shifting toward the dynamics associated with fiscal dominance, including persistent inflation, rising debt burdens, and prolonged real economic weakness.

7. Welfare Analysis

7.1. Welfare Criterion

To quantitatively evaluate the macroeconomic performance of the coexistence regime relative to the benchmark active monetary and passive fiscal policy framework (Regime M), we specify a welfare criterion grounded in the household's unconditional expected lifetime utility. Following the standard macroeconomic framework set out in [Schmitt-Grohe and Uribe \(2007\)](#), aggregate welfare, W_t , is defined recursively as:

$$W_t = U(c_t, N_t) + \beta E_t[W_{t+1}] \quad (43)$$

where the period utility function $U(c_t, N_t)$ assumes a Constant Relative Risk Aversion (CRRA) specification:

$$U(c_t, N_t) = \frac{c_t^{1-\sigma}}{1-\sigma} - \chi \frac{N_t^{1+\eta}}{1+\eta} \quad (44)$$

Within this calibration framework, the inverse of the intertemporal elasticity of substitution is set strictly greater than unity ($\sigma = 2$). Accurately computing the CE welfare transitions under CRRA utility requires decomposing aggregate welfare into its underlying consumption and labor disutility components, denoted as W_t^c and W_t^n , respectively:

$$W_t^c = \frac{c_t^{1-\sigma}}{1-\sigma} + \beta E_t[W_{t+1}^c] \quad (45)$$

and

$$W_t^n = \chi \frac{N_t^{1+\eta}}{1+\eta} + \beta E_t[W_{t+1}^n] \quad (46)$$

Aggregate welfare is therefore expressed as the net difference of these components, $W_t = W_t^c - W_t^n$.

The consumption equivalent, denoted by λ , provides an economically intuitive metric for these welfare comparisons. Specifically, λ defines the constant fractional change in baseline consumption required in every period under the benchmark regime (Regime M) to equate the household's unconditional expected welfare to that achieved under an alternative policy configuration (such as Regime F or the coexistence regime).

Formally, λ satisfies the following equality:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(c_t^M (1 + \lambda))^{1-\sigma}}{1-\sigma} - \chi \frac{(N_t^M)^{1+\eta}}{1+\eta} \right] = W^{Alt} \quad (47)$$

where W^{Alt} denotes the unconditional expected welfare generated by the alternative regime. Because the scale factor λ applies exclusively to the utility derived from consumption, it cannot be factored directly out of the net aggregate utility function. Isolating the consumption factor yields:

$$(1 + \lambda)^{1-\sigma} E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(c_t^M)^{1-\sigma}}{1-\sigma} \right] - E_0 \sum_{t=0}^{\infty} \beta^t \left[\chi \frac{(N_t^M)^{1+\eta}}{1+\eta} \right] = W^{Alt} \quad (48)$$

By substituting the recursively defined benchmark welfare components into Equation (48), the relationship simplifies to:

$$(1 + \lambda)^{1-\sigma} W_c^M - W_n^M = W^{Alt} \quad (49)$$

Solving algebraically for λ provides the exact analytical expression for the consumption equivalent under CRRA utility:

$$\lambda = \left(\frac{W^{Alt} + W_n^M}{W_c^M} \right)^{\frac{1}{1-\sigma}} - 1 \quad (50)$$

A positive value for λ indicates a strict welfare gain relative to Regime M, whereas a negative value reflects a comparative welfare loss. To execute this quantitative exercise, the theoretical moments of the defined welfare components are evaluated using a second-order Taylor approximation around the non-stochastic steady state. To prevent distortion of the unconditional means from explosive higher-order terms during the stochastic simulation, the standard pruning algorithm is applied.

7.2. Welfare Results

Table 2: Welfare Comparison Across Policy Regimes

Policy Regime	Unconditional Welfare ($E[W]$)	Consumption Equivalent (λ)
M	-186.4368	Benchmark
F	-189.4789	-2.2754%
Coexistence	-187.0767	-0.4874%

Notes: The consumption equivalent (λ) represents the constant percentage change in baseline Regime M consumption required to equate unconditional expected welfare under the alternative regime. A negative value denotes a welfare loss relative to the benchmark.

The outcomes of our welfare evaluation are detailed in Table 2. These findings confirm that Regime M serves as the superior benchmark among the evaluated frameworks. Within this foundational setup, an active monetary policy works alongside a passive fiscal approach to firmly anchor inflation expectations while simultaneously limiting early distortions to real interest rates. Keeping intertemporal debt fluctuations from disrupting price determinacy allows this framework to smooth both labour supply and household consumption throughout the business cycle. This mechanism ultimately generates the maximum unconditional expected welfare.

We observe a 2.28% welfare reduction for Regime F relative to the Regime M baseline. This alternative framework features active fiscal management combined with passive monetary policy. Consequently, the government depends on inflation rather than primary fiscal surpluses to achieve public debt stabilisation. When a contractionary shock occurs, the subsequent growth in unbacked obligations triggers extreme volatility in inflation. This instability interacts with Calvo price rigidities to create substantial distortions across relative prices. Such distortions severely compromise aggregate production efficiency and force households to endure significant utility losses due to suboptimal labour decisions and highly irregular consumption patterns.

Finally, our computations demonstrate that the coexistence framework thoroughly outperforms Regime F. This intermediate setup reduces the welfare penalty to just 0.49% against our primary benchmark. Within this blended environment, fiscal and monetary policymakers cooperate to neutralise a specific fraction of public obligations via inflation while managing the remainder using traditional fiscal surpluses. Segmenting the debt in this manner functions as a macroeconomic buffer that stops aggregate debt accumulation from becoming explosive and significantly mutes the required inflationary reaction. Despite these stabilisation benefits, the coexistence regime falls short of the optimal benchmark established by Regime M due to the distortionary effects of taxation. Specifically, the framework necessitates an increased tax on labour, which substantially amplifies the overall disutility of working. Even though households enjoy a higher welfare level from consumption under the coexistence regime compared to Regime M, this specific advantage is completely offset by the severe welfare penalty induced by the heightened labour disutility. Consequently, the net welfare state is driven down to a level strictly below the primary baseline.

8. Sensitivity Analysis: The Coexistence Regime

This section encapsulates the sensitivity analysis which is the contrast of the responses of the main macroeconomic variables following a 1% contractionary monetary policy shock under the coexistence regime. This analysis pertains to shocks to two parameters of our model: zeta and theta, which govern the size of unbacked public debt, and price stickiness, respectively. Section 8.1 considers the differentials in the main variables arising from changes in the size of the monetary policy shock that influences unbacked public debt. Similarly, Section 8.2 compares the responses of the same variables under flexible prices relative to sticky prices.

8.1. Varying unbacked public debt

Figure 2 illustrates the impulse response to a 1% contractionary monetary policy shock under the coexistence regime for three different alternative values of zeta, ζ , where ζ captures the proportion of the monetary policy shock that is transmitted to the unbacked component of public debt. A higher value of ζ therefore implies that a larger portion of the increase in public debt is stabilised through future inflation rather than through expected fiscal adjustment. This comparative analysis contributes towards identifying the policy mix between fiscal and monetary authorities that is most effective in containing public debt while preserving price stability, particularly during periods when fiscal pressures are elevated and available fiscal space is constrained.

Following the shock, debt service costs rise in all three scenarios as higher interest rates increase the cost of servicing outstanding government liabilities. However, the increase is largest when $\zeta = 0.25$, and smallest when $\zeta = 0.75$. As a result, the debt-to-GDP ratio rises most sharply when only a small portion of the shock affects unbacked debt, peaking at nearly 3% above steady state for $\zeta = 0.25$, compared with around 2% for $\zeta = 0.50$ and roughly 1% for $\zeta = 0.75$.

The inflation response highlights the key difference across scenarios. When $\zeta = 0.25$, inflation falls on impact and remains relatively subdued, indicating that debt stabilisation relies primarily on fiscal adjustment. Consistent with this, the primary surplus increases strongly, reaching approximately 3.5% of GDP. By contrast, when $\zeta = 0.75$, inflation rises persistently following monetary tightening, peaking at around 0.5 percentage points above steady state, while the primary surplus increases only modestly to about 1% of GDP. In this case, private agents expect a larger share of public debt to be stabilised through a higher price level rather than future taxation, which strengthens inflation expectations and reduces the need for fiscal consolidation.

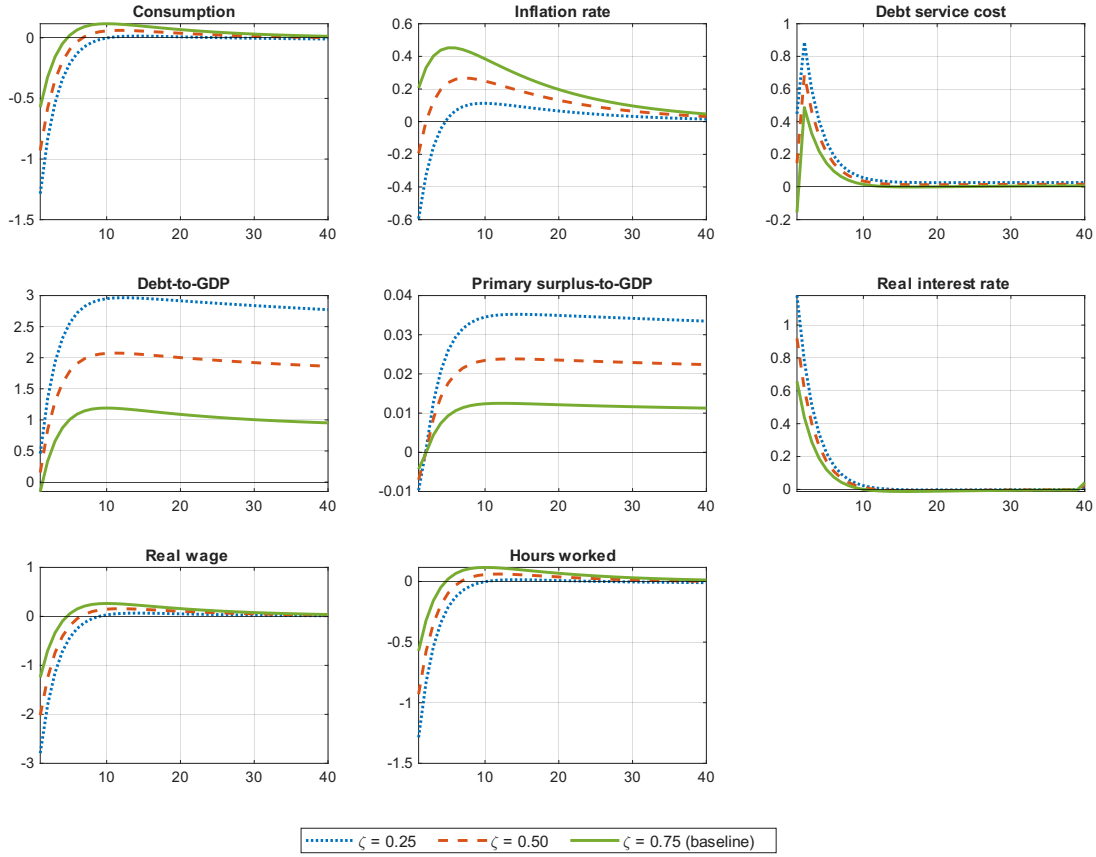


Figure 2: Impulse Responses to a 1% contractionary monetary policy shock under coexistence regime with varying sizes of ζ i.e monetary policy shock affecting unbacked public debt. $\zeta = 0.25$ (blue dotted line); $\zeta = 0.5$ (red dashed line); $\zeta = 0.75$ (green solid line). The variables are expressed as percentage deviations from steady state, except for the fiscal variables, which are expressed as linear deviations from steady state.

These differences are also reflected in the real interest rate. Although the real rate rises sharply in all cases, the increase is largest when $\zeta = 0.25$ and smallest when $\zeta = 0.75$. When inflation responds more strongly, it offsets a greater portion of the nominal rate increase, resulting in a milder rise in the ex post real interest rate.

The responses of consumption, real wages, and hours worked mirror these dynamics. All three variables decline on impact, but the contraction is substantially larger when $\zeta = 0.25$. Under $\zeta = 0.75$, the increase in inflation and the smaller rise in real interest rates cushion the decline in private demand and labour market activity, leading to a faster recovery. By contrast, when debt is predominantly backed by future fiscal surpluses, the sharper increase in real rates and stronger fiscal adjustment generate a deeper and more persistent contraction.

Overall, the results confirm that the macroeconomic effects of monetary tightening depend critically on how the associated increase in public debt is expected to be stabilised. Higher

values of ζ shift debt adjustment toward the inflation channel, resulting in a smaller increase in debt and a more muted decline in real activity, but at the cost of persistently higher inflation. Lower values of ζ place greater reliance on primary surpluses, which keeps inflation better anchored but leads to larger debt accumulation and a more pronounced economic contraction. This comparison helps identify the fiscal-monetary policy mix that is most effective in stabilising public debt while preserving price stability, particularly when fiscal space is limited.

8.2. Sensitivity to price stickiness

To assess how nominal rigidities shape the transmission of monetary policy under partial fiscal backing, we compare the impulse responses to a 1% contractionary monetary shock across three Calvo parameters: fully flexible prices ($\theta = 0$), moderate stickiness ($\theta = 0.5$), and high stickiness ($\theta = 0.75$, baseline). The results, as illustrated in Figure 3 reveal a fundamental monetary-fiscal trade-off: the degree of price flexibility determines whether the economy stabilizes public debt primarily through immediate price-level adjustment or through the fiscal inflation channel, with direct consequences for real activity and debt sustainability.

The most striking result is the divergence in inflation dynamics across price-setting regimes. Under flexible prices ($\theta = 0$), the contractionary shock produces an immediate and sharp deflation of approximately 1.5 percentage points. With all firms able to reprice instantly, the nominal tightening passes directly into lower prices. Conversely, under sticky prices ($\theta = 0.75$), inflation rises by approximately 0.75 percentage points on impact. This sign reversal is a hallmark of the Fiscal Theory of the Price Level (FTPL) in the coexistence regime: because prices cannot adjust immediately, private agents anticipate that the unbacked portion of the increased debt stock will ultimately be stabilized through the inflation tax. This expectation generates a positive wealth effect that boosts aggregate demand and inflation expectations. The intermediate case ($\theta = 0.5$) exhibits a moderate positive inflation response.

The behaviour of the real interest rate follows directly from the Fisher equation and the inflation dynamics described above. Under flexible prices, the immediate deflation nearly fully offsets the rise in the nominal policy rate, leaving the ex-ante real interest rate near zero throughout the horizon. Under sticky prices, the inability to deflate immediately means the nominal tightening translates into a substantial rise in the real rate (peaking at 0.6 pp on impact), which then gradually decays as inflation rises and the nominal rate reverts to the mean. Contrary to intuition, the real recession is deepest under flexible prices, not sticky prices. Under $\theta = 0$, consumption, real wages, and hours worked all fall

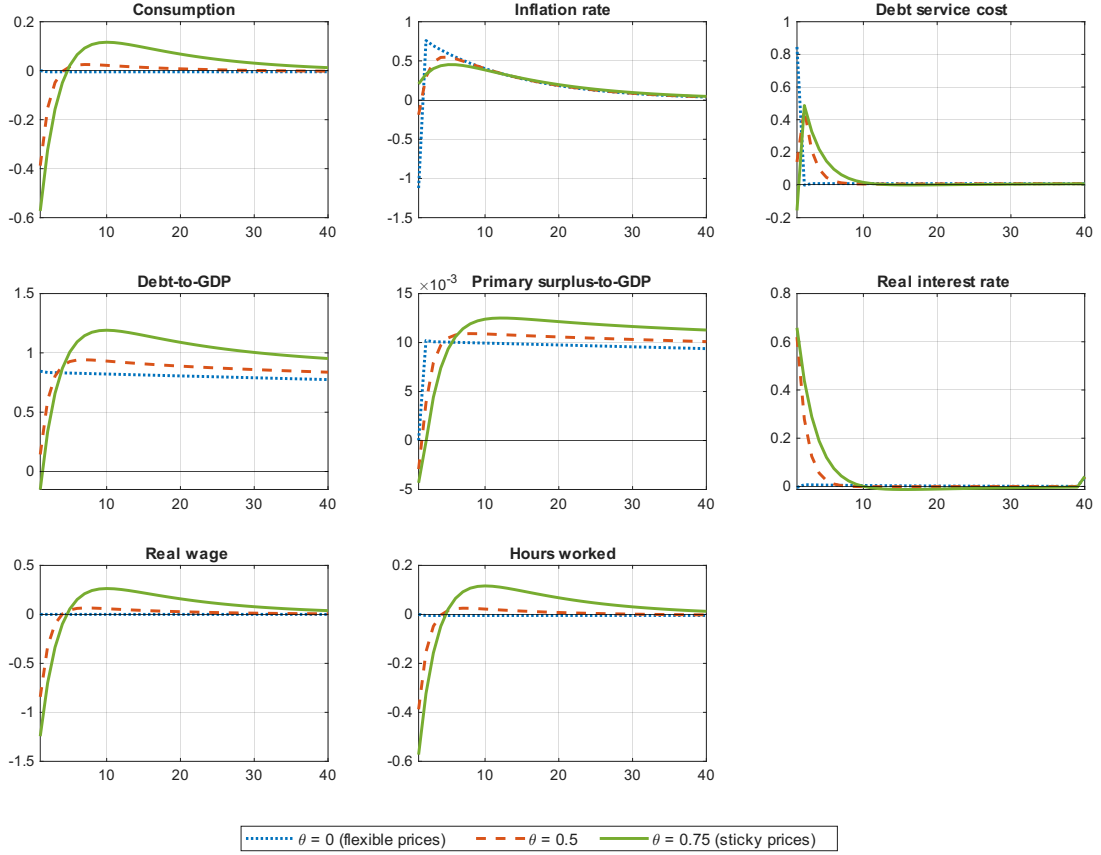


Figure 3: Impulse Responses to a 1% contractionary monetary policy shock under coexistence regime with varying degrees of price stickiness represented by θ . $\theta = 0$ represents flexible prices (blue dotted line); $\theta = 0.5$ (red dashed line); $\theta = 0.75$ for sticky prices (green solid line). The variables are expressed as percentage deviations from steady state, except for the fiscal variables, which are expressed as linear deviations from steady state.

sharply on impact, driven by the combination of a sharp real deflation and the immediate fiscal adjustment. The recovery is fast, with a brief positive overshoot as prices and quantities converge to the equilibrium. Under $\theta = 0.75$, the initial decline in all three variables is much smaller which arises because the positive wealth effect from anticipated inflation-based debt stabilisation partially offsets the contractionary demand effect, and the real interest rate, while elevated, rises by far less than in the flexible-price case in the very short run once the inflation channel is accounted for. The subsequent positive overshoot is also more pronounced under $\theta = 0.75$, reflecting this sustained wealth effect.

The degree of price stickiness critically alters the debt stabilization mechanism. Under flexible prices, the debt-to-GDP ratio jumps immediately to approximately 0.8% of GDP and then stabilizes, as the rapid price-level adjustment erodes the real value of nominal liabilities without requiring prolonged increased in primary surpluses. Correspondingly, the primary surplus adjusts instantaneously and remains flat under flexible prices. The debt service cost spike is modest and quickly dissipates. In contrast, under sticky prices, debt

accumulates more dynamically, peaking near 1.2% of GDP. Because the inflation cannot immediately absorb the fiscal shock, the real burden of debt persists, amplified by higher real interest rates. This drives a significantly larger spike in debt service costs, nearly three times the size in the flexible-price case. The primary surplus adjusts more gradually but converges to a comparable long-run level, as the fiscal rule tightens to stabilize the larger accumulated debt. The delayed fiscal adjustment underscores that sticky prices introduce a heavier reliance on future surplus generation, whereas flexible prices permit immediate balance sheet repair.

The θ sensitivity analysis highlights a critical monetary-fiscal trade-off in the coexistence regime. Greater price flexibility enables rapid inflation adjustment, anchoring price stability and minimizing debt accumulation, but at the cost of a sharper, albeit short-lived, real economic contraction. Greater price stickiness weakens the transmission of nominal tightening into real deflation, allowing the fiscal dominance channel to push inflation upward and cushion the real economy. However, this comes at the expense of higher debt service costs, more pronounced debt accumulation, and a prolonged elevation of real interest rates. These findings demonstrate that the degree of nominal rigidities are not merely a friction that slows price adjustment; they fundamentally reshape the monetary-fiscal transmission mechanism by altering the relative efficacy of the inflation tax versus fiscal consolidation in stabilizing partially unbacked public debt.

References

- Aruoba, S.B., Cuba-Borda, P. and Schorfheide, F. 2018. Macroeconomic dynamics near the ZLB: A tale of two countries. *The Review of Economic Studies*, 85 (1): 87–118.
- Bernanke, B.S. and Mishkin, F.S. 1997. Inflation targeting: a new framework for monetary policy? *Journal of Economic perspectives*, 11 (2): 97–116.
- Bianchi, F., Faccini, R. and Melosi, L. 2023. A fiscal theory of persistent inflation. *The Quarterly Journal of Economics*, 138 (4): 2127–2179.
- Bianchi, F., Melosi, L. and Rogantini-Picco, A. 2025. Monetary/fiscal policy mix and inflation dynamics. *Research Handbook on Inflation*, 125–139.
- Chen, X., Leeper, E.M. and Leith, C. 2022. Strategic interactions in US monetary and fiscal policies. *Quantitative Economics*, 13 (2): 593–628.
- Davig, T. and Leeper, E.M. 2011. Monetary–fiscal policy interactions and fiscal stimulus. *European Economic Review*, 55 (2): 211–227.
- Galí, J. 2015. *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and its Applications*. 2nd edn. Princeton University Press.
- Hofmann, B., Manea, C. and Mojon, B. 2024. Targeted taylor rules: monetary policy responses to demand-and supply-driven inflation. *BIS Quarterly Review*, 241 .
- Khah, A.M. and Ahmad, M. 2025. Investigating the dynamic nexus between fiscal deficit and inflation: empirical evidence from brics economies. *International Journal of Emerging Markets*, 20 (11): 4714–4738.
- Leeper, E.M. 1991. Equilibria under ‘active’and ‘passive’monetary and fiscal policies. *Journal of monetary Economics*, 27 (1): 129–147.
- Leeper, E.M. and Leith, C. 2016. Understanding inflation as a joint monetary–fiscal phenomenon. In: *Handbook of macroeconomics* 2, 2305–2415. Elsevier.
- Leeper, E.M., Traum, N. and Walker, T.B. 2017. Clearing up the fiscal multiplier morass. *American Economic Review*, 107 (8): 2409–2454.
- Loewald, C., Steinbach, R. and Rakgalakane, J. 2025. *Less Risk and More Reward: Revising South Africa’s Inflation Target*. Economic Research and Statistics Department, South African Reserve Bank.
- Rawdanowicz, Ł., Turban, S., Haas, J., Crowe, D. and Millot, V. 2021. Constraints and

- demands on public finances: Considerations of resilient fiscal policy. *OECD Economic Department Working Papers*, (1694): 1–71.
- Rotemberg, J.J. and Woodford, M. 1997. An optimization-based econometric framework for the evaluation of monetary policy. *NBER Macroeconomics Annual*, 12 : 297–346.
- Schmitt-Grohe, S. and Uribe, M. 2007 September. Optimal simple and implementable monetary and fiscal rules. *Journal of Monetary Economics*, 54 (6): 1702–1725.
- Sims, C.A. 1994. A simple model for study of the determination of the price level and the interaction of monetary and fiscal policy. *Economic theory*, 4 (3): 381–399.
- Smets, F. and Wouters, R. 2003. An estimated dynamic stochastic general equilibrium model of the Euro area. *Journal of the European Economic Association*, 1 (5): 1123–1175.
- Smets, F. and Wouters, R. 2007. Shocks and frictions in US business cycles: A Bayesian DSGE approach. *American Economic Review*, 97 (3): 586–606.
- Smets, F. and Wouters, R. 2024. Fiscal backing, inflation and US business cycles. In: *Federal Reserve Bank of San Francisco Working Paper*. Federal Reserve Bank of San Francisco.
- Taylor, J.B. 1999. A historical analysis of monetary policy rules. In: *Monetary policy rules*, 319–348. University of Chicago Press.
- Wansleben, L. 2023. *The rise of central banks: State power in financial capitalism*. Harvard University Press.
- Weber, P. and Feld, L.P. 2025. Government debt and inflation: A cross-country investigation of the fiscal theory of the price level. Tech. Rep., Walter Eucken Institut eV.

A. Complete System of Nonlinear Equilibrium Equations

This appendix presents the complete set of nonlinear equilibrium conditions for both the actual economy and the shadow economy. Together, these two systems fully characterise the model's equilibrium. Variables carrying a superscript f belong to the shadow economy; all remaining variables belong to the actual economy. A variable written without a time subscript denotes its non-stochastic steady-state value.

A.1 Actual Economy

The actual economy consists of 17 endogenous variables: $y_t, N_t, c_t, R_t, w_t, b_t, Z_t, s_t, \tau_t, \tau_t^n, g_t, \Pi_t, P_t^*, d_t, g_{1,t}, g_{2,t}, mc_t, dsc_t$, driven by an i.i.d. monetary policy shock ε_t^R and an AR(1) technology shock ξ_t^z (with $\rho_z = 0.9$).

(1) Euler equation (bond FOC):

$$\frac{1}{R_t} = \beta \mathbb{E}_t \left[\left(\frac{c_t}{c_{t+1}} \right)^\sigma \frac{1}{\Pi_{t+1}} \right]. \quad (\text{A.1})$$

(2) Labour supply (labour FOC):

$$\chi N_t^\eta c_t^\sigma = \frac{(1 - \tau_t^n)}{(1 + \tau^c)} w_t. \quad (\text{A.2})$$

(3) Real marginal cost:

$$mc_t = \frac{w_t}{Z_t}. \quad (\text{A.3})$$

(4) Production function (with price dispersion):

$$Z_t N_t = y_t d_t. \quad (\text{A.4})$$

(5) Technology shock:

$$\ln Z_t = \rho_z \ln Z_{t-1} + \xi_t^z. \quad (\text{A.5})$$

(6) Aggregate price index (Calvo):

$$1 = \theta \Pi_t^{v-1} + (1 - \theta)(P_t^*)^{1-v}. \quad (\text{A.6})$$

(7) Price dispersion:

$$d_t = \theta \Pi_t^v d_{t-1} + (1 - \theta)(P_t^*)^{-v}. \quad (\text{A.7})$$

(8) Optimal reset price ratio:

$$P_t^* = \frac{v}{v-1} \cdot \frac{g_{1,t}}{g_{2,t}}. \quad (\text{A.8})$$

(9) Auxiliary price-setting variable $g_{1,t}$:

$$g_{1,t} = y_t m c_t c_t^{-\sigma} + \beta \theta \mathbb{E}_t \left[\Pi_{t+1}^v g_{1,t+1} \right]. \quad (\text{A.9})$$

(10) Auxiliary price-setting variable $g_{2,t}$:

$$g_{2,t} = y_t c_t^{-\sigma} + \beta \theta \mathbb{E}_t \left[\Pi_{t+1}^{v-1} g_{2,t+1} \right]. \quad (\text{A.10})$$

(11) Goods market clearing:

$$y_t = c_t + g_t. \quad (\text{A.11})$$

(12) Real tax revenue:

$$\tau_t = \tau^c c_t + \tau_t^n w_t N_t. \quad (\text{A.12})$$

(13) Primary surplus:

$$s_t = \tau_t - g_t. \quad (\text{A.13})$$

(14) Debt dynamics:

$$b_t = \frac{R_{t-1}}{\Pi_t} b_{t-1} - s_t. \quad (\text{A.14})$$

(15) Debt service cost:

$$dsc_t = \left(\frac{R_{t-1}}{\Pi_t} - 1 \right) b_{t-1}. \quad (\text{A.15})$$

(16) Fiscal rule (coexistence regime):

$$\frac{s_t}{y_t} = \left(\frac{s}{y} \right) \left(\frac{b_{t-1}/(R_{t-1}y_{t-1})}{b_{t-1}^f/(R_{t-1}y_{t-1})} \right)^{\delta^{PF}} \left(\frac{b_{t-1}^f/(R_{t-1}y_{t-1})}{b/(Ry)} \right)^{\delta^{AF}}. \quad (\text{A.16})$$

(17) Monetary rule (coexistence regime):

$$\frac{1}{R_t} = \frac{1}{R} \left(\frac{1/R_{t-1}}{1/R} \right)^{\rho_R} \left[\left(\frac{1/\Pi_t}{1/\Pi_t^f} \right)^{\alpha_\pi^{AM}} \left(\frac{1/\Pi_t^f}{1/\Pi} \right)^{\alpha_\pi^{PM}} \right]^{1-\rho_R} e^{\varepsilon_t^R}. \quad (\text{A.17})$$

A.2 Shadow Economy

The shadow economy supplements the actual economy and tracks unbacked public debt b_t^f , fiscal inflation Π_t^f , and the shadow debt service cost dsc_t^f . It consists of the same 18 equations as the actual economy, with all variables replaced by their f -superscripted counterparts, and with the following modified policy rules and constraints.

(S1) Shadow fiscal rule (Regime F):

$$\frac{s_t^f}{y_t^f} = \left(\frac{s}{y}\right) \left(\frac{b_{t-1}^f/(R_{t-1}^f y_{t-1}^f)}{b/(Ry)}\right)^{\delta^{AF}}. \quad (\text{A.19})$$

(S2) Shadow monetary rule (passive, Regime F):

$$\frac{1}{R_t^f} = \frac{1}{R} \left(\frac{1/R_{t-1}^f}{1/R}\right)^{\rho_R} \left(\frac{1/\Pi_t^f}{1/\Pi}\right)^{\alpha_\pi^{PM}(1-\rho_R)} e^{\zeta \varepsilon_t^R}. \quad (\text{A.20})$$

(S3) Shadow debt service cost:

$$dsc_t^f = \left(\frac{R_{t-1}^f}{\Pi_t^f} - 1\right) b_{t-1}^f. \quad (\text{A.21})$$

A.3 Standard Regimes M and F

Under the two standard regimes, the model economy collapses to a single economy governed by the following policy rules in place of (A.16) and (A.17).

Standard fiscal rule:

$$\frac{s_t}{y_t} = \left(\frac{s}{y}\right) \left(\frac{b_{t-1}/(R_{t-1} y_{t-1})}{b/(Ry)}\right)^{\delta}. \quad (\text{A.22})$$

Standard monetary rule:

$$\frac{1}{R_t} = \frac{1}{R} \left(\frac{1/R_{t-1}}{1/R}\right)^{\rho_R} \left(\frac{1/\Pi_t}{1/\Pi}\right)^{\alpha_\pi(1-\rho_R)} e^{\varepsilon_t^R}. \quad (\text{A.23})$$

Regime M is obtained when $\alpha_\pi \beta > 1$ and $\beta^{-1} - \delta < 1$ (active monetary, passive fiscal). **Regime F** is obtained when $\alpha_\pi \beta < 1$ and $\beta^{-1} - \delta > 1$ (passive monetary, active fiscal). All remaining equilibrium conditions continue to hold unchanged under both standard regimes.

B. Steady State Conditions

Table B.1: Steady-state conditions

Equation	Expression
Fisher equation	$R = \frac{1}{\beta}$
Labour supply	$w = \frac{(1 + \tau^c)}{(1 - \tau^n)} \chi N^\eta c^\sigma$
Real marginal cost	$mc = \frac{v - 1}{v}$
Real wage	$w = \frac{v - 1}{v}$
Production function	$y = N$
Price dispersion	$d = 1$
Government budget constraint	$s = (\beta^{-1} - 1) b$
Debt service cost	$dsc = \left(\frac{R}{\Pi} - 1\right) b$
Primary surplus-to-GDP ratio	$\frac{s}{y} = (\beta^{-1} - 1) \frac{b}{y}$
Goods market clearing	$y = c + g$
Labour disutility parameter	$\chi = \frac{(1 - \tau^n)}{(1 + \tau^c)} \cdot \frac{(v - 1)/v}{N^\eta c^\sigma}$

Notes: Variables without time subscripts denote steady-state values. Gross aggregate inflation is normalised to unity ($\Pi = 1$) and the technology level to $Z = 1$. The consumption tax rate τ^c is constant and the steady-state labour income tax rate τ^n is calibrated to satisfy the government budget constraint.

C. Data and Sources

This appendix documents the empirical time series used to calibrate the model's steady-state ratios (Section 4) and to compute the business cycle moments reported in Section 5. All series are obtained from the Federal Reserve Bank of St. Louis Economic Database (FRED, <https://fred.stlouisfed.org/>). The sample spans 1982:Q1 to 2025:Q3 at quarterly frequency. Monthly series are aggregated to quarterly frequency by simple within-quarter averaging. Nominal series are converted to real terms by deflating with the GDP deflator (FRED: GDPDEF), and all real aggregate quantities are expressed in per-capita terms by dividing by the OECD working-age population for the United States, ages 15–64 (FRED: LFWA64TTUSM647S); for the latter series, the quarterly data are available only through September 2025, which dictates the terminal date of the sample.

Table C.1 lists each variable together with its FRED mnemonic and the construction used

to map the raw data into the model counterpart. The consumption tax rate $\tau^c = 0.023$ is a calibrated parameter rather than a data series and follows [Bianchi *et al.* \(2023\)](#); the steady-state labour income tax rate τ^n is recovered residually from the steady-state government budget constraint and is therefore not constructed from data.

Table C.1: Empirical series used in calibration and moment computation

Model variable	FRED ID	Series and construction
Output (y_t)	GDP	<i>Gross Domestic Product (nominal)</i> . Deflated by GDPDEF and divided by working-age population to obtain real per-capita output; used to compute b/y and g/y in calibration and as the cyclical output measure in moment computation.
Consumption (c_t)	PCECC96	<i>Real Personal Consumption Expenditures</i> . Already in real (chained) terms; divided by working-age population for per-capita measure.
Public debt (b_t)	GFDEBTN	<i>Federal Debt: Total Public Debt (nominal, end-of-period)</i> . Deflated by GDPDEF and divided by working-age population for real per-capita debt; nominal-over-nominal b/y used directly to set the calibrated steady-state ratio.
Govt. expenditure (g_t)	GCE	<i>Government Consumption Expenditures and Gross Investment (nominal)</i> . Nominal-over-nominal g/y used directly to set the calibrated steady-state ratio.
Inflation (Π_t)	CPIAUCSL	<i>Consumer Price Index for All Urban Consumers: All Items</i> . Quarterly index level constructed as the within-quarter mean of the monthly series; gross inflation computed as $\Pi_t = \text{CPI}_t / \text{CPI}_{t-1}$.
Interest rate (R_t)	FEDFUNDS	<i>Federal Funds Effective Rate (annualised, monthly)</i> . Quarterly value taken as within-quarter mean; converted to gross quarterly rate as $R_t = 1 + \text{FEDFUNDS}_t / 400$. Sample mean (3.9% annualised) corroborates the calibrated steady-state value $R = 1/\beta = 1.01$.
<i>Supporting (auxiliary) series</i>		
GDP deflator	GDPDEF	<i>Gross Domestic Product: Implicit Price Deflator</i> . Used to deflate GDP and GFDEBTN to obtain real output and real public debt.
Working-age pop.	LFWA64TTUSM647S	<i>Working Age Population: Aged 15–64, All Persons (OECD)</i> . Monthly series aggregated to quarterly by within-quarter averaging; used as the denominator for per-capita conversion of all aggregate real quantities.

Notes: All series cover 1982:Q1–2025:Q3 at quarterly frequency. Variables entering the business cycle moment computation in Section 1 are first expressed in real per-capita terms, log-transformed, and detrended using the Hodrick–Prescott filter with smoothing parameter $\lambda = 1600$; the gross inflation rate Π_t and the gross nominal interest rate R_t are HP-detrended without prior log-transformation, consistent with their gross-form treatment in the model. Calibrated steady-state ratios b/y and g/y are computed as the simple sample means of the corresponding nominal-over-nominal ratios.

D. Simulation Code

This appendix contains the Dynare source code used to compute the impulse response functions and welfare metrics for the three policy regimes examined in this study.

D.1. Impulse Response Functions (IRFs)

For the impulse response simulations in section 6 , we use a first-order approximation (`order=1`). Since the model's steady state is stable and determinate, a linear solution adequately captures the qualitative transmission of a one-standard-deviation shock. Utilising the first-order approximation isolates the core economic mechanisms and ensures visual transparency in the plots by abstracting from the additional curvature of higher-order terms, which, most importantly, do not alter the directional or persistence patterns of the responses. This is evident in figure D.1, being of `order=2`, but having near identical impulse response functions as those in figure 1.

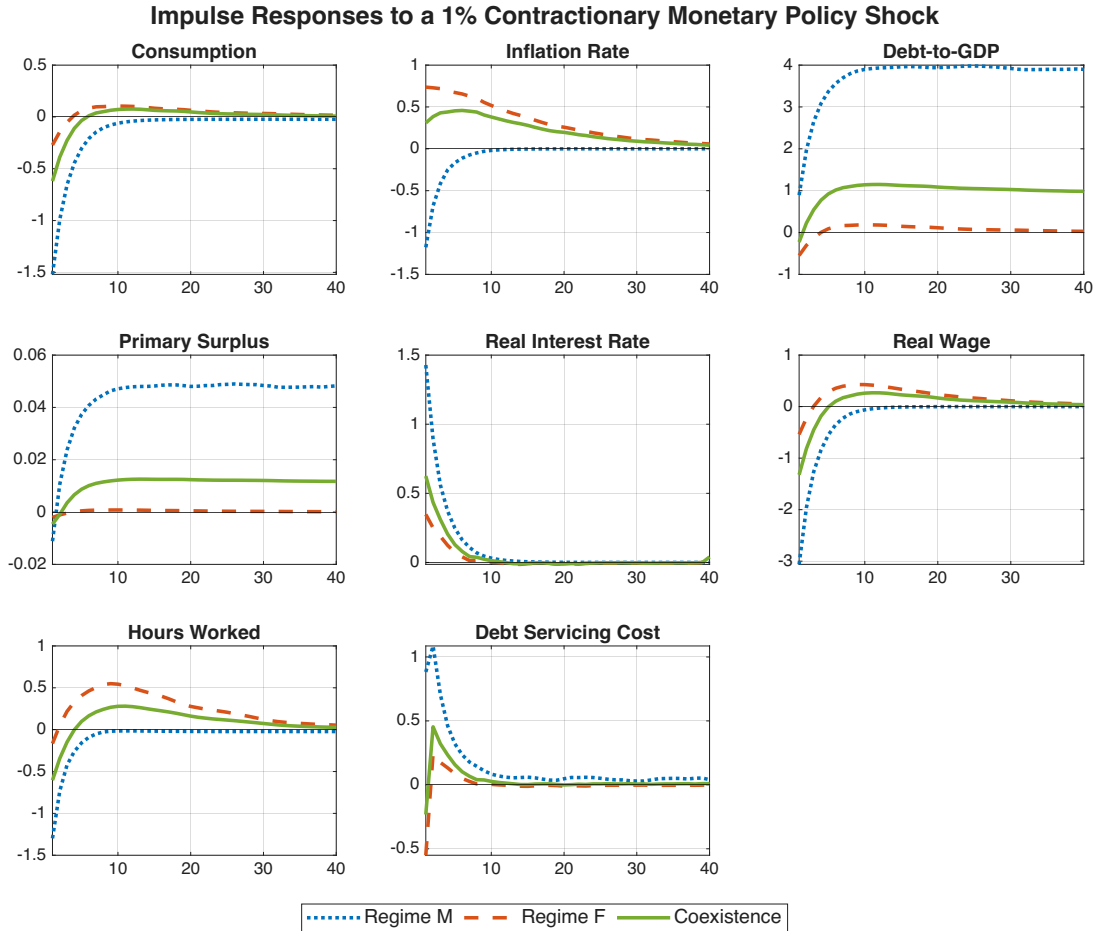


Figure D.1: Impulse Responses to a Contractionary Monetary Policy Shock

D.1.1 Coexistence Regime

```
%-----
% Macroeconomics 871      Group Project
% Authors:   Diaz Sangeve      24991422
%           Sihlelelwe Nzima    23854936
%           Tshulani Khosa      25852094
% Coexistence Regime
%-----

var
    y      N      c      R      w      b      Z      s      tau      tau_n      g
    Pi      P_star      d      g1      g2      mc      dsc
    b_f      Pi_f      b_m      Pi_m
    y_f      N_f      c_f      R_f      w_f
    Z_f      s_f      tau_f      tau_n_f      g_f
    P_star_f      d_f      g1_f      g2_f      mc_f      dsc_f
;

varexo eps_R eps_A;

%-----
% Parameter declaration
%-----

parameters
    beta sigma eta chi theta v
    alpha_AM_pi alpha_PM_pi rho_R
    delta_PF delta_AF
    zeta tau_c
    g_ss b_ss y_ss c_ss Pi_ss R_ss
    w_ss N_ss s_ss tau_ss tau_n_ss Z_ss dsc_ss
;

beta      = 0.99;
sigma     = 2;
eta       = 1;
v         = 9;
theta     = 0.75;

b_ss      = 0.746;
```

```

g_ss      = 0.230;
tau_c     = 0.023;

alpha_AM_pi = 1.5;
alpha_PM_pi = 0.5;
rho_R      = 0.85;
delta_PF   = 1.2;
delta_AF    = 0;
zeta       = 0.75;

% --- Standard Baseline Steady State ---
Pi_ss     = 1;
R_ss      = 1 / beta;

y_ss      = 1;
c_ss      = y_ss - g_ss;
w_ss      = (v - 1) / v;
N_ss      = y_ss;
Z_ss      = 1;

s_ss      = (R_ss / Pi_ss - 1) * b_ss;
tau_ss    = s_ss + g_ss;
tau_n_ss  = (tau_ss - tau_c * c_ss) / (w_ss * N_ss);
dsc_ss    = (R_ss / Pi_ss - 1) * b_ss;

chi = w_ss * (1 - tau_n_ss) / ((1 + tau_c) * (c_ss^sigma) * (N_ss
    ^eta));

model;

%-----
% 1. ACTUAL ECONOMY
%-----

% Households & Firms
c^(-sigma) = beta * c(+1)^(-sigma) * (R / Pi(+1));
chi * N^eta * c^sigma = w * (1 - tau_n) / (1 + tau_c);
mc = w / Z;
y = c + g;
Z * N = y * d;
log(Z) = 0.9 * log(Z(-1)) + eps_A;

```

```

% Calvo Pricing
g1 = mc * y * c^(-sigma) + beta * theta * Pi(+1)^v * g1(+1);
g2 = y * c^(-sigma) + beta * theta * Pi(+1)^(v-1) * g2(+1);
g1 = P_star * g2 * ((v - 1) / v);
1 = theta * Pi^(v - 1) + (1 - theta) * P_star^(1 - v);
d = theta * Pi^v * d(-1) + (1 - theta) * P_star^(-v);

% Government & Debt Dynamics
g = g_ss;
tau = tau_n * w * N + tau_c * c;
s = tau - g;
b = (R(-1) / Pi) * b(-1) - s;
dsc = (R(-1) / Pi - 1) * b(-1);

b_m = b - b_f;
Pi_m = Pi / Pi_f;

% Fiscal Rule
s / y = (s_ss / y_ss)
        * ( (b(-1) / (R(-1) * y(-1))) / (b_f(-1) / (R(-1) * y
            (-1))) )^delta_PF
        * ( (b_f(-1) / (R(-1) * y(-1))) / (b_ss / (R_ss * y_ss)
            ) )^delta_AF;

% Monetary Rule
R / R_ss = (R(-1) / R_ss)^rho_R
          * ( (Pi_m / Pi_ss)^alpha_AM_pi * (Pi_f / Pi_ss)^
              alpha_PM_pi )^(1 - rho_R)
          * exp(eps_R);

%-----
% 2. SHADOW ECONOMY
%-----

% Households & Firms
c_f^(-sigma) = beta * c_f(+1)^(-sigma) * (R_f / Pi_f(+1));
chi * N_f^eta * c_f^sigma = w_f * (1 - tau_n_f) / (1 + tau_c)
;
mc_f = w_f / Z_f;

```

```

y_f = c_f + g_f;
Z_f * N_f = y_f * d_f;
log(Z_f) = 0.9 * log(Z_f(-1)) + eps_A;

% Calvo Pricing
g1_f = mc_f * y_f * c_f^(-sigma) + beta * theta * Pi_f(+1)^v
      * g1_f(+1);
g2_f = y_f * c_f^(-sigma) + beta * theta * Pi_f(+1)^(v-1) *
      g2_f(+1);
g1_f = P_star_f * g2_f * ((v - 1) / v);
1 = theta * Pi_f^(v - 1) + (1 - theta) * P_star_f^(1 - v);
d_f = theta * Pi_f^v * d_f(-1) + (1 - theta) * P_star_f^(-v);

% Government & Debt Dynamics
g_f = g_ss;
tau_f = tau_n_f * w_f * N_f + tau_c * c_f;
s_f = tau_f - g_f;
b_f = (R_f(-1) / Pi_f) * b_f(-1) - s_f;
dsc_f = (R_f(-1) / Pi_f - 1) * b_f(-1);

% Shadow Fiscal Rule
s_f / y_f = (s_ss / y_ss);

% Shadow Monetary Rule
R_f / R_ss = (R_f(-1) / R_ss)^rho_R
            * (Pi_f / Pi_ss)^((1 - rho_R) * alpha_PM_pi)
            * exp(zeta * eps_R);
end;

initval;

y      = y_ss;      c      = c_ss;      g      = g_ss;
b      = b_ss;      s      = s_ss;      R      = R_ss;
Pi     = Pi_ss;     N      = N_ss;     w      = w_ss;
mc     = w_ss;     P_star = 1;      d      = 1;
Z      = Z_ss;     tau_n   = tau_n_ss;  tau     = tau_ss;
dsc    = dsc_ss;

b_f    = b_ss;      b_m    = 0;
Pi_f   = Pi_ss;     Pi_m   = 1;
g1     = (mc * y * c^(-sigma)) / (1 - beta * theta);

```



```

g2      = (y * c^(-sigma)) / (1 - beta * theta);

y_f     = y_ss;          c_f     = c_ss;          g_f     = g_ss;
R_f     = R_ss;
N_f     = N_ss;          w_f     = w_ss;
Z_f     = Z_ss;          tau_n_f = tau_n_ss;
s_f     = s_ss;
tau_f   = s_f + g_ss;
mc_f    = w_ss;          P_star_f = 1;
d_f     = 1;
dsc_f   = dsc_ss;
g1_f    = g1;            g2_f    = g2;
end;

steady;
check;

shocks;
    var eps_R; stderr 0.01;
end;

stoch_simul(order=1, irf=40) y c Pi b R s w N dsc b_m Pi_m b_f
    Pi_f dsc_f;

```

D.1.2 Regime M

```

%-----
% Regime M      Active Monetary / Passive Fiscal
%-----

var
    y      N      c      R      w      b      Z      s      tau      tau_n      g
    Pi     P_star  d      g1     g2     mc     dsc
;

varexo eps_R eps_A;

parameters
    beta sigma eta chi theta v
    alpha_pi rho_R

```

```

    delta tau_c
    g_ss b_ss y_ss c_ss Pi_ss R_ss
    w_ss N_ss s_ss tau_ss tau_n_ss Z_ss dsc_ss
;

beta      = 0.99;
sigma     = 2;
eta       = 1;
v         = 9;
theta     = 0.75;

b_ss      = 0.746;
g_ss      = 0.230;
tau_c     = 0.023;

alpha_pi  = 1.5;
rho_R     = 0.85;
delta     = 1.2;

% --- Standard Baseline Steady State ---
Pi_ss     = 1;
R_ss      = 1 / beta;

y_ss      = 1;
c_ss      = y_ss - g_ss;
w_ss      = (v - 1) / v;
N_ss      = y_ss;
Z_ss      = 1;

s_ss      = (R_ss / Pi_ss - 1) * b_ss;
tau_ss    = s_ss + g_ss;
tau_n_ss  = (tau_ss - tau_c * c_ss) / (w_ss * N_ss);
dsc_ss    = (R_ss / Pi_ss - 1) * b_ss;

chi = w_ss * (1 - tau_n_ss) / ((1 + tau_c) * (c_ss^sigma) * (N_ss
    ^eta));

model;
    % Households & Firms
    c^(-sigma) = beta * c(+1)^(-sigma) * (R / Pi(+1));

```

```

chi * N^eta * c^sigma = w * (1 - tau_n) / (1 + tau_c);
mc = w / Z;
y = c + g;
Z * N = y * d;
log(Z) = 0.9 * log(Z(-1)) + eps_A;

% Calvo Pricing
g1 = mc * y * c^(-sigma) + beta * theta * Pi(+1)^v * g1(+1);
g2 = y * c^(-sigma) + beta * theta * Pi(+1)^(v-1) * g2(+1);
g1 = P_star * g2 * ((v - 1) / v);
1 = theta * Pi^(v - 1) + (1 - theta) * P_star^(1 - v);
d = theta * Pi^v * d(-1) + (1 - theta) * P_star^(-v);

% Government & Debt Dynamics
g = g_ss;
tau = tau_n * w * N + tau_c * c;
s = tau - g;
b = (R(-1) / Pi) * b(-1) - s;
dsc = (R(-1) / Pi - 1) * b(-1);

% Fiscal rule (Passive)
s / y = (s_ss / y_ss)
        * ( (b(-1) / (R(-1) * y(-1))) / (b_ss / (R_ss * y_ss))
            )^delta;

% Monetary rule (Active)
R / R_ss = (R(-1) / R_ss)^rho_R * (Pi / Pi_ss)^((1 - rho_R) *
        alpha_pi) * exp(eps_R);
end;

initval;
y      = y_ss;      c      = c_ss;      g      = g_ss;
b      = b_ss;      s      = s_ss;      R      = R_ss;
Pi     = Pi_ss;     N      = N_ss;     w      = w_ss;
mc     = w_ss;     P_star = 1;     d      = 1;
Z      = Z_ss;     tau_n   = tau_n_ss;   tau    = tau_ss;
dsc    = dsc_ss;
g1     = (mc * y * c^(-sigma)) / (1 - beta * theta);
g2     = (y * c^(-sigma)) / (1 - beta * theta);
end;

```

```

steady;
check;

shocks;
    var eps_R; stderr 0.01;
end;

stoch_simul(order=1, irf=40) y c Pi b R s w N dsc;

```

D.1.3 Regime F

```

%-----
% Regime F      Passive Monetary / Active Fiscal
%-----

var
    y      N      c      R      w      b      Z      s      tau      tau_n      g
    Pi     P_star  d      g1     g2     mc     dsc
;

varexo eps_R eps_A;

parameters
    beta sigma eta chi theta v
    alpha_pi rho_R
    delta tau_c
    g_ss b_ss y_ss c_ss Pi_ss R_ss
    w_ss N_ss s_ss tau_ss tau_n_ss Z_ss dsc_ss
;

beta      = 0.99;
sigma     = 2;
eta       = 1;
v         = 9;
theta     = 0.75;

b_ss      = 0.746;
g_ss      = 0.230;
tau_c     = 0.023;

```

```

alpha_pi = 0.5;
rho_R    = 0.85;
delta    = 0;

% --- Standard Baseline Steady State ---
Pi_ss    = 1;
R_ss     = 1 / beta;

y_ss     = 1;
c_ss     = y_ss - g_ss;
w_ss     = (v - 1) / v;
N_ss     = y_ss;
Z_ss     = 1;

s_ss     = (R_ss / Pi_ss - 1) * b_ss;
tau_ss   = s_ss + g_ss;
tau_n_ss = (tau_ss - tau_c * c_ss) / (w_ss * N_ss);
dsc_ss   = (R_ss / Pi_ss - 1) * b_ss;

chi = w_ss * (1 - tau_n_ss) / ((1 + tau_c) * (c_ss^sigma) * (N_ss
    ^eta));

model;

% Households & Firms
c^(-sigma) = beta * c(+1)^(-sigma) * (R / Pi(+1));
chi * N^eta * c^sigma = w * (1 - tau_n) / (1 + tau_c);
mc = w / Z;
y = c + g;
Z * N = y * d;
log(Z) = 0.9 * log(Z(-1)) + eps_A;

% Calvo Pricing
g1 = mc * y * c^(-sigma) + beta * theta * Pi(+1)^v * g1(+1);
g2 = y * c^(-sigma) + beta * theta * Pi(+1)^(v-1) * g2(+1);
g1 = P_star * g2 * ((v - 1) / v);
1 = theta * Pi^(v - 1) + (1 - theta) * P_star^(1 - v);
d = theta * Pi^v * d(-1) + (1 - theta) * P_star^(-v);

% Government & Debt Dynamics

```

```

g = g_ss;
tau = tau_n * w * N + tau_c * c;
s = tau - g;
b = (R(-1) / Pi) * b(-1) - s;
dsc = (R(-1) / Pi - 1) * b(-1);

% Fiscal rule (Active)
s / y = (s_ss / y_ss)
        * ( (b(-1) / (R(-1) * y(-1))) / (b_ss / (R_ss * y_ss))
            )^delta;

% Monetary rule (Passive)
R / R_ss = (R(-1) / R_ss)^rho_R * (Pi / Pi_ss)^((1 - rho_R) *
        alpha_pi) * exp(eps_R);
end;

initval;
y      = y_ss;      c      = c_ss;      g      = g_ss;
b      = b_ss;      s      = s_ss;      R      = R_ss;
Pi     = Pi_ss;     N      = N_ss;     w      = w_ss;
mc     = w_ss;     P_star = 1;     d      = 1;
Z      = Z_ss;     tau_n   = tau_n_ss;   tau    = tau_ss;
dsc    = dsc_ss;
g1     = (mc * y * c^(-sigma)) / (1 - beta * theta);
g2     = (y * c^(-sigma)) / (1 - beta * theta);
end;

steady;
check;

shocks;
    var eps_R; stderr 0.01;
end;

stoch_simul(order=1, irf=40)
y c Pi b R s w N dsc;

```

D.1.4 Plotting of IRFs

```

%-----

```

```

% Macroeconomics 871 - Group Project
% IRF Extraction and Plotting Script
%-----

clear; clc; close all;

% 1. Load the IRF structures from the Dynare Output subfolders
load('Regime_M/Output/Regime_M_results.mat', 'oo_');
irf_M = oo_.irfs;

load('Regime_F/Output/Regime_F_results.mat', 'oo_');
irf_F = oo_.irfs;

load('Coexistence/Output/Coexistence_results.mat', 'oo_');
irf_C = oo_.irfs;

% 2. Horizon
horizon = 40;
t = 1:horizon;

% 3. Variable names
vars = {'c_eps_R', 'Pi_eps_R', 'b_eps_R', 's_eps_R', 'R_eps_R',
        'w_eps_R', 'N_eps_R', 'dsc_eps_R'};
titles = {'Consumption', 'Inflation Rate', 'Debt-to-GDP', ...
          'Primary Surplus', 'Real Interest Rate', 'Real Wage',
          ...
          'Hours Worked', 'Debt Servicing Cost'};

% 4. Extract nominal IRFs safely
yM = cell(1,8); yF = cell(1,8); yC = cell(1,8);
for i = 1:8
    yM{i} = extract_irf(irf_M, vars{i}, horizon);
    yF{i} = extract_irf(irf_F, vars{i}, horizon);
    yC{i} = extract_irf(irf_C, vars{i}, horizon);
end

% 5. Compute real interest rate
pi_M = yM{2}; R_M = yM{5};
pi_F = yF{2}; R_F = yF{5};
pi_C = yC{2}; R_C = yC{5};

```

```

real_rate_M = R_M - [pi_M(2:end); 0];
real_rate_F = R_F - [pi_F(2:end); 0];
real_rate_C = R_C - [pi_C(2:end); 0];

% Overwrite the 5th variable with real rate
yM{5} = real_rate_M;
yF{5} = real_rate_F;
yC{5} = real_rate_C;

% 6. Scale factors:
% Consumption, Real Wage, Hours Worked : % deviation from SS
% Inflation Rate : percentage points
% Debt-to-GDP, Primary Surplus, Debt Servicing Cost : % of GDP
% Real Interest Rate : percentage points
scale = 100;

for i = 1:8
    yM{i} = yM{i} * scale;
    yF{i} = yF{i} * scale;
    yC{i} = yC{i} * scale;
end

% 7. Create figure
figure('Name', 'IRFs to a Contractionary Monetary Policy Shock',
    ...
    'Color', 'w', 'Position', [100, 100, 1000, 750]);

% 8. Plot all 8 panels
for i = 1:8
    subplot(3, 3, i);
    hold on;

    p1 = plot(t, yM{i}, ':', 'Color', [0 0.4470 0.7410], '
        LineWidth', 2);
    p2 = plot(t, yF{i}, '--', 'Color', [0.8500 0.3250 0.0980], '
        LineWidth', 2);
    p3 = plot(t, yC{i}, '-.', 'Color', [0.4660 0.6740 0.1880], '
        LineWidth', 2);

```



```

    title(titles{i}, 'FontSize', 11, 'FontWeight', 'bold');
    xlim([1 horizon]);
    yline(0, 'k-', 'LineWidth', 0.5, 'HandleVisibility', 'off');
    grid on;
    box on;
    hold off;
end

sgtitle('Impulse Responses to a 1% Contractionary Monetary Policy
    Shock', ...
        'FontSize', 14, 'FontWeight', 'bold');

% 9. Global legend at bottom
Lgnd = legend([p1, p2, p3], 'Regime M', 'Regime F', 'Coexistence'
    , ...
        'Orientation', 'horizontal', 'FontSize', 11);
Lgnd.Position(1) = 0.5 - Lgnd.Position(3)/2; % Center
    horizontally
Lgnd.Position(2) = 0.03; % Fix near bottom

%-----
% HELPER FUNCTION
%-----

function y = extract_irf(irf_struct, var_name, expected_length)
    if isfield(irf_struct, var_name) && ~isempty(irf_struct.(
        var_name))
        y = irf_struct.(var_name);
        if size(y,1)==1, y = y'; end
        if length(y) ~= expected_length
            warning('%s length %d, expected %d. Trimming/padding.
                ', ...
                    var_name, length(y), expected_length);
            if length(y) > expected_length
                y = y(1:expected_length);
            else
                y(end+1:expected_length) = 0;
            end
        end
    else
        y = zeros(expected_length, 1);
    end
end

```

```

        end
    end
end

```

D.2. Welfare Analysis

The welfare analysis, however, requires a second-order approximation (`order=2`). Because the household's utility function is non-linear, a first-order solution is insufficient as its inherent certainty equivalence fails to capture the welfare costs of volatility. Solving to the second order incorporates the necessary curvature of the decision rules, allowing for a comparison of expected lifetime utility across regimes based on both the means and variances of consumption and hours worked.

All second-order simulations are executed using the pruning algorithm (`pruning`). This prevents the explosive sample paths that arise from higher-order terms, particularly when the model is subjected to persistent shocks. By discarding the explosive components of the quadratic solution, pruning ensures numerical stability and keeps the simulated series within an economically sensible range.

D.2.1 Coexistence Regime

```

%-----
% Coexistence Regime WELFARE ANALYSIS
%-----

var
    y      N      c      R      w      b      Z      s      tau      tau_n      g
    Pi     P_star  d      g1     g2     mc     dsc
    b_f    Pi_f    b_m    Pi_m
    y_f    N_f    c_f    R_f    w_f
    Z_f    s_f    tau_f    tau_n_f  g_f
    P_star_f d_f    g1_f    g2_f    mc_f    dsc_f
    W      W_c    W_n
;

varexo eps_R eps_A;

%-----
% Parameter declaration
%-----

parameters

```

```

    beta sigma eta chi theta v
    alpha_AM_pi alpha_PM_pi rho_R
    delta_PF delta_AF
    zeta tau_c
    g_ss b_ss y_ss c_ss Pi_ss R_ss
    w_ss N_ss s_ss tau_ss tau_n_ss Z_ss dsc_ss
;

beta      = 0.99;
sigma     = 2;
eta       = 1;
v         = 9;
theta     = 0.75;

b_ss      = 0.746;
g_ss      = 0.230;
tau_c     = 0.023;

alpha_AM_pi = 1.5;
alpha_PM_pi = 0.5;
rho_R      = 0.85;
delta_PF   = 1.2;
delta_AF   = 0;
zeta       = 0.75;

% --- Standard Baseline Steady State ---
Pi_ss     = 1;
R_ss      = 1 / beta;

y_ss      = 1;
c_ss      = y_ss - g_ss;
w_ss      = (v - 1) / v;
N_ss      = y_ss;
Z_ss      = 1;

s_ss      = (R_ss / Pi_ss - 1) * b_ss;
tau_ss    = s_ss + g_ss;
tau_n_ss  = (tau_ss - tau_c * c_ss) / (w_ss * N_ss);
dsc_ss    = (R_ss / Pi_ss - 1) * b_ss;

```

```

chi = w_ss * (1 - tau_n_ss) / ((1 + tau_c) * (c_ss^sigma) * (N_ss
    ^eta));

model;
%-----
% 1. ACTUAL ECONOMY
%-----

% Households & Firms
c^(-sigma) = beta * c(+1)^(-sigma) * (R / Pi(+1));
chi * N^eta * c^sigma = w * (1 - tau_n) / (1 + tau_c);
mc = w / Z;
y = c + g;
Z * N = y * d;
log(Z) = 0.9 * log(Z(-1)) + eps_A;

% Calvo Pricing
g1 = mc * y * c^(-sigma) + beta * theta * Pi(+1)^v * g1(+1);
g2 = y * c^(-sigma) + beta * theta * Pi(+1)^(v-1) * g2(+1);
g1 = P_star * g2 * ((v - 1) / v);
1 = theta * Pi^(v - 1) + (1 - theta) * P_star^(1 - v);
d = theta * Pi^v * d(-1) + (1 - theta) * P_star^(-v);

% Government & Debt Dynamics
g = g_ss;
tau = tau_n * w * N + tau_c * c;
s = tau - g;
b = (R(-1) / Pi) * b(-1) - s;
dsc = (R(-1) / Pi - 1) * b(-1);

b_m = b - b_f;
Pi_m = Pi / Pi_f;

% Fiscal Rule
s / y = (s_ss / y_ss)
    * ( (b(-1) / (R(-1) * y(-1))) / (b_f(-1) / (R(-1) * y
        (-1))) )^delta_PF
    * ( (b_f(-1) / (R(-1) * y(-1))) / (b_ss / (R_ss * y_ss)
        ) )^delta_AF;

```

```

% Monetary Rule
R / R_ss = (R(-1) / R_ss)^rho_R
          * ( (Pi_m / Pi_ss)^alpha_AM_pi * (Pi_f / Pi_ss)^
              alpha_PM_pi )^(1 - rho_R)
          * exp(eps_R);

%-----
% 2. SHADOW ECONOMY
%-----

% Households & Firms
c_f^(-sigma) = beta * c_f(+1)^(-sigma) * (R_f / Pi_f(+1));
chi * N_f^eta * c_f^sigma = w_f * (1 - tau_n_f) / (1 + tau_c)
;
mc_f = w_f / Z_f;
y_f = c_f + g_f;
Z_f * N_f = y_f * d_f;
log(Z_f) = 0.9 * log(Z_f(-1)) + eps_A;

% Calvo Pricing
g1_f = mc_f * y_f * c_f^(-sigma) + beta * theta * Pi_f(+1)^v
      * g1_f(+1);
g2_f = y_f * c_f^(-sigma) + beta * theta * Pi_f(+1)^(v-1) *
      g2_f(+1);
g1_f = P_star_f * g2_f * ((v - 1) / v);
1 = theta * Pi_f^(v - 1) + (1 - theta) * P_star_f^(1 - v);
d_f = theta * Pi_f^v * d_f(-1) + (1 - theta) * P_star_f^(-v);

% Government & Debt Dynamics
g_f = g_ss;
tau_f = tau_n_f * w_f * N_f + tau_c * c_f;
s_f = tau_f - g_f;
b_f = (R_f(-1) / Pi_f) * b_f(-1) - s_f;
dsc_f = (R_f(-1) / Pi_f - 1) * b_f(-1);

% Shadow Fiscal Rule
s_f / y_f = (s_ss / y_ss);

% Shadow Monetary Rule
R_f / R_ss = (R_f(-1) / R_ss)^rho_R

```

```

        * (Pi_f / Pi_ss)^((1 - rho_R) * alpha_PM_pi)
        * exp(zeta * eps_R);

%
-----

% WELFARE (CRRA Specification)
%
-----

W_c = (c^(1-sigma))/(1-sigma) + beta*W_c(+1);
W_n = chi*(N^(1+eta))/(1+eta) + beta*W_n(+1);
W    = W_c - W_n;
end;

initval;
y      = y_ss;      c      = c_ss;      g      = g_ss;
b      = b_ss;      s      = s_ss;      R      = R_ss;
Pi     = Pi_ss;     N      = N_ss;      w      = w_ss;
mc     = w_ss;      P_star = 1;      d      = 1;
Z      = Z_ss;      tau_n   = tau_n_ss;   tau     = tau_ss;
dsc    = dsc_ss;

b_f    = b_ss;      b_m    = 0;
Pi_f   = Pi_ss;     Pi_m   = 1;
g1     = (mc * y * c^(-sigma)) / (1 - beta * theta);
g2     = (y * c^(-sigma)) / (1 - beta * theta);

y_f    = y_ss;      c_f    = c_ss;      g_f    = g_ss;
R_f    = R_ss;
N_f    = N_ss;      w_f    = w_ss;
Z_f    = Z_ss;      tau_n_f = tau_n_ss;
s_f    = s_ss;
tau_f  = s_f + g_ss;
mc_f   = w_ss;      P_star_f = 1;
d_f    = 1;
dsc_f  = dsc_ss;
g1_f   = g1;      g2_f   = g2;

% Steady-state welfare: W_ss = U_ss / (1 - beta)

```

```

        W_c = ( (c_ss^(1-sigma))/(1-sigma) ) / (1 - beta);
        W_n = ( chi*(N_ss^(1+eta))/(1+eta) ) / (1 - beta);
        W    = W_c - W_n;
end;

steady;
check;

shocks;
    var eps_R; stderr 0.01;
end;

stoch_simul(order=2, irf=40, pruning) y c Pi b R s w N dsc b_m
    Pi_m b_f Pi_f dsc_f W_c W_n W;

```

D.2.2 Regime M

```

%-----
% Regime M WELFARE ANALYSIS
%-----

var
    y      N      c      R      w      b      Z      s      tau      tau_n      g
    Pi      P_star  d      g1      g2      mc      dsc
    W      W_c      W_n
;

varexo eps_R eps_A;

parameters
    beta sigma eta chi theta v
    alpha_pi rho_R
    delta tau_c
    g_ss b_ss y_ss c_ss Pi_ss R_ss
    w_ss N_ss s_ss tau_ss tau_n_ss Z_ss dsc_ss
;

beta      = 0.99;
sigma     = 2;
eta       = 1;

```

```

v          = 9;
theta      = 0.75;

b_ss       = 0.746;
g_ss       = 0.230;
tau_c      = 0.023;

alpha_pi   = 1.5;           % Active monetary
rho_R      = 0.85;
delta      = 1.2;           % Passive fiscal

% --- Standard Baseline Steady State ---
Pi_ss      = 1;
R_ss       = 1 / beta;

y_ss       = 1;
c_ss       = y_ss - g_ss;
w_ss       = (v - 1) / v;
N_ss       = y_ss;
Z_ss       = 1;

s_ss       = (R_ss / Pi_ss - 1) * b_ss;
tau_ss     = s_ss + g_ss;
tau_n_ss   = (tau_ss - tau_c * c_ss) / (w_ss * N_ss);
dsc_ss     = (R_ss / Pi_ss - 1) * b_ss;

chi = w_ss * (1 - tau_n_ss) / ((1 + tau_c) * (c_ss^sigma) * (N_ss
    ^eta));

model;

% Households & Firms
c^(-sigma) = beta * c(+1)^(-sigma) * (R / Pi(+1));
chi * N^eta * c^sigma = w * (1 - tau_n) / (1 + tau_c);
mc = w / Z;
y = c + g;
Z * N = y * d;
log(Z) = 0.9 * log(Z(-1)) + eps_A;

% Calvo Pricing
g1 = mc * y * c^(-sigma) + beta * theta * Pi(+1)^v * g1(+1);

```



```

g2 = y * c^(-sigma) + beta * theta * Pi(+1)^(v-1) * g2(+1);
g1 = P_star * g2 * ((v - 1) / v);
1 = theta * Pi^(v - 1) + (1 - theta) * P_star^(1 - v);
d = theta * Pi^v * d(-1) + (1 - theta) * P_star^(-v);

% Government & Debt Dynamics
g = g_ss;
tau = tau_n * w * N + tau_c * c;
s = tau - g;
b = (R(-1) / Pi) * b(-1) - s;
dsc = (R(-1) / Pi - 1) * b(-1);

% Fiscal rule (Passive)
s / y = (s_ss / y_ss)
        * ( (b(-1) / (R(-1) * y(-1))) / (b_ss / (R_ss * y_ss))
            )^delta;

% Monetary rule (Active)
R / R_ss = (R(-1) / R_ss)^rho_R * (Pi / Pi_ss)^((1 - rho_R) *
        alpha_pi) * exp(eps_R);

%
-----

% WELFARE (CRRA Specification)
%
-----

W_c = (c^(1-sigma))/(1-sigma) + beta*W_c(+1);
W_n = chi*(N^(1+eta))/(1+eta) + beta*W_n(+1);
W    = W_c - W_n;
end;

initval;
y      = y_ss;      c      = c_ss;      g      = g_ss;
b      = b_ss;      s      = s_ss;      R      = R_ss;
Pi     = Pi_ss;     N      = N_ss;     w      = w_ss;
mc     = w_ss;      P_star = 1;      d      = 1;
Z      = Z_ss;      tau_n   = tau_n_ss;   tau     = tau_ss;
dsc    = dsc_ss;

```

```

g1      = (mc * y * c^(-sigma)) / (1 - beta * theta);
g2      = (y * c^(-sigma)) / (1 - beta * theta);

W_c = ( (c_ss^(1-sigma))/(1-sigma) ) / (1 - beta);
W_n = ( chi*(N_ss^(1+eta))/(1+eta) ) / (1 - beta);
W      = W_c - W_n;
end;

steady;
check;

shocks;
    var eps_R; stderr 0.01;
end;

stoch_simul(order=2, irf=40, pruning) y c Pi b R s w N dsc W_c
    W_n W;

```

D.2.3 Regime F

```

%-----
% Regime F WELFARE ANALYSIS
%-----

var
    y      N      c      R      w      b      Z      s      tau      tau_n      g
    Pi     P_star  d      g1     g2     mc     dsc
    W      W_c     W_n
;

varexo eps_R eps_A;

parameters
    beta sigma eta chi theta v
    alpha_pi rho_R
    delta tau_c
    g_ss b_ss y_ss c_ss Pi_ss R_ss
    w_ss N_ss s_ss tau_ss tau_n_ss Z_ss dsc_ss
;

```

```

beta      = 0.99;
sigma     = 2;
eta       = 1;
v         = 9;
theta     = 0.75;

b_ss      = 0.746;
g_ss      = 0.230;
tau_c     = 0.023;

alpha_pi  = 0.5;
rho_R     = 0.85;
delta     = 0;

% --- Standard Baseline Steady State ---
Pi_ss     = 1;
R_ss      = 1 / beta;

y_ss      = 1;
c_ss      = y_ss - g_ss;
w_ss      = (v - 1) / v;
N_ss      = y_ss;
Z_ss      = 1;

s_ss      = (R_ss / Pi_ss - 1) * b_ss;
tau_ss    = s_ss + g_ss;
tau_n_ss  = (tau_ss - tau_c * c_ss) / (w_ss * N_ss);
dsc_ss    = (R_ss / Pi_ss - 1) * b_ss;

chi = w_ss * (1 - tau_n_ss) / ((1 + tau_c) * (c_ss^sigma) * (N_ss
    ^eta));

model;
    % Households & Firms
    c^(-sigma) = beta * c(+1)^(-sigma) * (R / Pi(+1));
    chi * N^eta * c^sigma = w * (1 - tau_n) / (1 + tau_c);
    mc = w / Z;
    y = c + g;
    Z * N = y * d;
    log(Z) = 0.9 * log(Z(-1)) + eps_A;

```

```

% Calvo Pricing
g1 = mc * y * c^(-sigma) + beta * theta * Pi(+1)^v * g1(+1);
g2 = y * c^(-sigma) + beta * theta * Pi(+1)^(v-1) * g2(+1);
g1 = P_star * g2 * ((v - 1) / v);
1 = theta * Pi^(v - 1) + (1 - theta) * P_star^(1 - v);
d = theta * Pi^v * d(-1) + (1 - theta) * P_star^(-v);

% Government & Debt Dynamics
g = g_ss;
tau = tau_n * w * N + tau_c * c;
s = tau - g;
b = (R(-1) / Pi) * b(-1) - s;
dsc = (R(-1) / Pi - 1) * b(-1);

% Fiscal rule (Active)
s / y = (s_ss / y_ss)
        * ( (b(-1) / (R(-1) * y(-1))) / (b_ss / (R_ss * y_ss))
            )^delta;

% Monetary rule (Passive)
R / R_ss = (R(-1) / R_ss)^rho_R * (Pi / Pi_ss)^((1 - rho_R) *
        alpha_pi) * exp(eps_R);

%
-----

% WELFARE (CRRA Specification)
%
-----

W_c = (c^(1-sigma))/(1-sigma) + beta*W_c(+1);
W_n = chi*(N^(1+eta))/(1+eta) + beta*W_n(+1);
W    = W_c - W_n;
end;

initval;

y      = y_ss;      c      = c_ss;      g      = g_ss;
b      = b_ss;      s      = s_ss;      R      = R_ss;
Pi     = Pi_ss;     N      = N_ss;     w      = w_ss;

```

```

mc      = w_ss;          P_star = 1;          d      = 1;
Z       = Z_ss;          tau_n   = tau_n_ss;    tau     = tau_ss;
dsc     = dsc_ss;
g1      = (mc * y * c^(-sigma)) / (1 - beta * theta);
g2      = (y * c^(-sigma)) / (1 - beta * theta);

W_c = ( (c_ss^(1-sigma))/(1-sigma) ) / (1 - beta);
W_n = ( chi*(N_ss^(1+eta))/(1+eta) ) / (1 - beta);
W     = W_c - W_n;
end;

steady;
check;

shocks;
    var eps_R; stderr 0.01;
end;

stoch_simul(order=2, irf=40, pruning) y c Pi b R s w N dsc W_c
    W_n W;

```

D.2.4 Welfare results

```

%-----
% Welfare Analysis Driver
%
% Runs Regime M, Regime F, and the Coexistence regime in turn at
% order = 2, extracts the unconditional mean of welfare (E[W])
% from each,
% and computes the Consumption Equivalent (CE) of Regime F and
% the
% Coexistence regime relative to Regime M.
%
% CE_i = exp[ (1 - beta) * (W_i - W^M) ] - 1
%-----

clear all; close all; clc;

%-----
% 1. Regime M (benchmark)

```

```

%-----
fprintf('\n===== Running Regime M =====\n');
dynare Regime_M_welfare noclearall nolog

% oo_.mean is indexed by the variables in stoch_simul: ... W_c
    W_n W
W_c_M = oo_.mean(end-2);
W_n_M = oo_.mean(end-1);
W_M    = oo_.mean(end);
sigma_val = M_.params(strcmp(cellstr(M_.param_names), 'sigma'));

%-----
% 2. Regime F
%-----
fprintf('\n===== Running Regime F =====\n');
dynare Regime_F_welfare noclearall nolog
W_F = oo_.mean(end);

%-----
% 3. Coexistence regime
%-----
fprintf('\n===== Running Coexistence regime =====\n');
dynare Coexistence_welfare noclearall nolog
W_C = oo_.mean(end);

%-----
% 4. Compute consumption-equivalent welfare (CRRA exact formula)
%    lambda = ( (W_new + W_n_old) / W_c_old )^(1/(1-sigma)) - 1
%-----
CE_F = ( (W_F + W_n_M) / W_c_M )^(1 / (1 - sigma_val)) - 1;
CE_C = ( (W_C + W_n_M) / W_c_M )^(1 / (1 - sigma_val)) - 1;

CE_F_percent = CE_F * 100;
CE_C_percent = CE_C * 100;

%-----
% 5. Display results
%-----
fprintf('\n');

```

```

fprintf('
=====
n');
fprintf('    WELFARE ANALYSIS - Consumption Equivalent (Regime M =
    benchmark)\n');
fprintf('
=====
n');
fprintf('    Unconditional welfare E[W]:\n');
fprintf('        Regime M                : %12.6f\n', W_M);
fprintf('        Regime F                : %12.6f\n', W_F);
fprintf('        Coexistence regime : %12.6f\n', W_C);
fprintf('
-----
n');
fprintf('    Consumption Equivalent (%%):\n');
fprintf('        Regime F                : %+9.4f %%\n', CE_F_percent);
fprintf('        Coexistence regime : %+9.4f %%\n', CE_C_percent);
fprintf('
=====
n');

%-----
% 6. Save results for use in the report
%-----

welfare_results = table( ...
    {'Regime M'; 'Regime F'; 'Coexistence'}, ...
    [W_M; W_F; W_C], ...
    [0; CE_F; CE_C], ...
    'VariableNames', {'Regime', 'Unconditional_Welfare', '
        CE_percent'} );
disp(welfare_results);
writetable(welfare_results, 'welfare_results.csv');
fprintf('\nResults written to welfare_results.csv\n');

```